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# *Algebra toward Algebraic Geometry*

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# Algebra toward Algebraic Geometry

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# Chapter 1

## Commutative Algebra

### 1.1 Some fundamental Results

Yang: To be completed

#### 1.1.1 Rings and modules

In the appendix and all the note, the “ring” is always commutative and with identity. We denote by  $\text{Spec } A$  the set of prime ideals of a ring  $A$ . We denote by  $\text{mSpec } A$  the set of maximal ideals of  $A$ . Let  $I \subset A$  be an ideal of  $A$ . We define

$$V(I) := \{\mathfrak{p} \in \text{Spec } A : I \subset \mathfrak{p}\}.$$

Let  $\mathfrak{a}, \mathfrak{b}$  be ideals of  $A$ . We define

$$(\mathfrak{a} : \mathfrak{b}) := \{a \in A : a\mathfrak{b} \subset \mathfrak{a}\}.$$

This is an ideal of  $A$ .

Let  $\text{rad}(A)$  be the Jacobian radical of  $A$ , i.e., the intersection of all maximal ideals of  $A$ . Let  $\text{nil}(A)$  be the nilradical of  $A$ , i.e., the ideal of  $A$  consisting of all nilpotent elements.

**Proposition 1.1.1.** Let  $A$  be a ring. Then we have

$$\text{nil}(A) = \bigcap_{\mathfrak{p} \in \text{Spec } A} \mathfrak{p}.$$

*Proof.* Yang: To be completed. □

**Proposition 1.1.2.** Let  $A$  be a ring,  $\mathfrak{p}, \mathfrak{p}_i$  prime ideals of  $A$  and  $\mathfrak{a}, \mathfrak{a}_i$  ideals of  $A$ .

- (a) Suppose  $\mathfrak{a} \subset \bigcup_{i=1}^n \mathfrak{p}_i$ . Then there exists  $i$  such that  $\mathfrak{a} \subset \mathfrak{p}_i$ .
- (b) Suppose  $\bigcap_{i=1}^n \mathfrak{a}_i \subset \mathfrak{p}$ . Then there exists  $i$  such that  $\mathfrak{a}_i \subset \mathfrak{p}$ .

*Proof.* Yang: To be completed. □

Let  $M$  be an  $A$ -module. We say that  $M$  is *finite* if there exists an exact sequence

$$A^n \rightarrow M \rightarrow 0.$$

We say that  $M$  is *finite presented* if there exists an exact sequence

$$A^m \rightarrow A^n \rightarrow M \rightarrow 0.$$

If  $A$  is a noetherian ring, then every finite  $A$ -module is finite presented.

**Definition 1.1.3.** Let  $A$  be a ring and  $M$  an  $A$ -module. The *support* of  $M$  is defined as

$$\text{Supp } M := \{\mathfrak{p} \in \text{Spec } A : M_{\mathfrak{p}} \neq 0\}.$$

The *annihilator* of  $M$  is defined as

$$\text{Ann } M := \{a \in A : aM = 0\}.$$

This is an ideal of  $A$ .

**Proposition 1.1.4.** Let  $A$  be a ring and  $M$  a finite  $A$ -module. Then  $\text{Supp } M = V(\text{Ann } M)$ . In particular,  $\text{Supp } M$  is a closed subset of  $\text{Spec } A$ .

*Proof.* Yang: To be completed. □

### 1.1.2 Localization

**Definition 1.1.5.** Let  $A$  be a ring and  $S \subset A$  a multiplicative subset, i.e.,  $1 \in S$  and  $s_1, s_2 \in S$  implies  $s_1 s_2 \in S$ . Let  $M$  be an  $A$ -module. The *localization* of  $M$  at  $S$  is defined as

$$S^{-1}M := M \times S / \sim,$$

where  $(m, s) \sim (n, t)$  if there exists  $u \in S$  such that  $u(tm - sn) = 0$ . We denote the equivalence class of  $(m, s)$  by  $\frac{m}{s}$  or  $m/s$ .

The localization  $S^{-1}A$  is still a ring and hence an  $A$ -algebra. The localization  $S^{-1}M$  is an  $S^{-1}A$ -module. If  $M = B$  is an  $A$ -algebra, then  $S^{-1}B$  is an  $S^{-1}A$ -algebra.

**Example 1.1.6.** Let  $A$  be a ring,  $\mathfrak{p}$  a prime ideal of  $A$  and  $M$  an  $A$ -module. Then  $S = A \setminus \mathfrak{p}$  is a multiplicative subset. The localization  $S^{-1}M$  is denoted by  $M_{\mathfrak{p}}$  and called the localization of  $M$  at  $\mathfrak{p}$ .

Let  $f \in A$  be an element. Then  $S = \{f^n : n \geq 0\}$  is a multiplicative subset. The localization  $S^{-1}M$  is denoted by  $M[1/f]$ .

**Proposition 1.1.7.** The natural map  $A \rightarrow S^{-1}A$  is injective if and only if  $S$  contains no zero divisors.

**Proposition 1.1.8.** Let  $A$  be a ring,  $S \subset A$  a multiplicative subset and  $M$  an  $A$ -module. Then we have a natural isomorphism of  $S^{-1}A$ -modules

$$S^{-1}M \cong M \otimes_A S^{-1}A.$$

**Proposition 1.1.9.** The localization  $S^{-1}A$  is a flat  $A$ -algebra.

### 1.1.3 Chain conditions

**Definition 1.1.10.** Let  $A$  be a ring. We say that  $A$  is *noetherian* (resp. *artinian*) if every ascending (resp. descending) chain of ideals of  $A$  stabilizes.

**Proposition 1.1.11.** Let  $A$  be a ring. The following are equivalent:

- (a)  $A$  is noetherian.
- (b) Every ideal of  $A$  is finitely generated.
- (c) Every non-empty set of ideals of  $A$  has a maximal element (with respect to inclusion).

*Proof.* Yang: To be completed. □

**Theorem 1.1.12** (Hilbert's Basis Theorem). If  $A$  is a noetherian ring, then  $A[x]$  is noetherian.

*Proof.* Yang: To be completed. □

**Remark 1.1.13.** By a similar argument replacing  $\deg f$  by  $\text{ord } f$ , we can show that if  $A$  is noetherian, then the formal power series ring  $A[[x]]$  is also noetherian.

### 1.1.4 Nakayama's Lemma

**Theorem 1.1.14** (Nakayama's Lemma). Let  $A$  be a ring and  $\mathfrak{M}$  be its Jacobi radical. Suppose  $M$  is a finitely generated  $A$ -module. If  $\mathfrak{a}M = M$  for  $\mathfrak{a} \subset \mathfrak{M}$ , then  $M = 0$ .

*Proof.* Suppose  $M$  is generated by  $x_1, \dots, x_n$ . Since  $M = \mathfrak{a}M$ , formally we have  $(x_1, \dots, x_n)^T = \Phi(x_1, \dots, x_n)^T$  for  $\Phi \in M_n(\mathfrak{a})$ . Then  $(\Phi - \text{id})(x_1, \dots, x_n)^T = 0$ . Note that  $\det(\Phi - \text{id}) = 1 + a$  for  $a \in \mathfrak{a} \subset \mathfrak{M}$ . Then  $\Phi - \text{id}$  is invertible and then  $M = 0$ . □

The finiteness of  $M$  is crucial in Nakayama's Lemma. The followings are counterexamples when  $M$  is not finite.

**Example 1.1.15.** Let  $\overline{\mathbb{Z}}$  be the ring of algebraic integers in  $\overline{\mathbb{Q}}$ . Choose a non-zero prime ideal  $\mathfrak{p}$  of  $\overline{\mathbb{Z}}$ . Then we have that  $\mathfrak{p}\overline{\mathbb{Z}}_{\mathfrak{p}} = \mathfrak{p}^2\overline{\mathbb{Z}}_{\mathfrak{p}}$ . Indeed, if  $a \in \mathfrak{p}\overline{\mathbb{Z}}_{\mathfrak{p}}$ , let  $b = \sqrt{a} \in \overline{\mathbb{Z}}_{\mathfrak{p}}$ . Then  $b^2 = a \in \mathfrak{p}\overline{\mathbb{Z}}_{\mathfrak{p}}$  and whence  $b \in \mathfrak{p}\overline{\mathbb{Z}}_{\mathfrak{p}}$  since  $\mathfrak{p}$  is prime. It follows that  $a = b^2 \in \mathfrak{p}^2\overline{\mathbb{Z}}_{\mathfrak{p}}$ .

**Example 1.1.16.** Let  $A = \mathbf{k}[x]$  and  $B = \mathbf{k}[t]$  with  $t^2 = x$ . Then  $B$  is a  $A$ -module. Let  $\mathfrak{m} = (x - 1) \subset A$  and  $\mathfrak{n} = (t - 1) \subset B$ . Then  $B_{\mathfrak{n}}$  is a  $A_{\mathfrak{m}}$ -module which is not finite. Note that  $\mathfrak{m}B_{\mathfrak{n}} = (t^2 - 1)B_{\mathfrak{n}} = (t - 1)(t + 1)B_{\mathfrak{n}} = \mathfrak{n}B_{\mathfrak{n}}$ . Then  $B_{\mathfrak{n}} \otimes_A \kappa(\mathfrak{m}) = \mathbf{k}$  is generated by 1. However,  $B_{\mathfrak{n}}$  is not generated by 1 as a  $A_{\mathfrak{m}}$ -module.

**Proposition 1.1.17** (Geometric form of Nakayama's Lemma). Let  $X = \text{Spec } A$  be an affine scheme,  $x \in X$  a closed point and  $\mathcal{F}$  a coherent sheaf on  $X$ . If  $a_1, \dots, a_k \in \mathcal{F}(X)$  generate  $\mathcal{F}|_x = \mathcal{F} \otimes \kappa(x)$ , then there is an open subset  $U \subset X$  such that  $a_i|_U$  generate  $\mathcal{F}(U)$ .

*Proof.* Yang: To be completed. □

**Corollary 1.1.18.** Let  $X$  be a scheme and  $\mathcal{F}$  a coherent sheaf on  $X$ . Then the function  $x \mapsto \dim_{\kappa(x)} \mathcal{F}|_x$  is upper semicontinuous.

*Proof.* Yang: To be completed. □

### 1.1.5 Nullstellensatz

Let  $\mathbf{k}$  be a field and  $\bar{\mathbf{k}}$  be its algebraic closure.

**Theorem 1.1.19** (Noether's Normalization Lemma). Let  $A$  be a  $\mathbf{k}$ -algebra of finite type. Then there is an injection  $\mathbf{k}[T_1, \dots, T_d] \hookrightarrow A$  such that  $A$  is finite over  $\mathbf{k}[T_1, \dots, T_d]$ .

**Remark 1.1.20.** Here  $A$  does not need to be integral. For example,

**Theorem 1.1.21** (Hilbert's Nullstellensatz). Let  $A$  be a  $\mathbf{k}$ -algebra of finite type.

- (a) If  $\mathfrak{m}$  is a maximal ideal of  $A$ , then  $A/\mathfrak{m}$  is a finite extension of  $\mathbf{k}$ .
- (b) Suppose that  $\mathbf{k}$  is algebraically closed and  $A = \mathbf{k}[x_1, \dots, x_n]/\mathfrak{a}$ . Then there is a bijection between the set of maximal ideals of  $A$  and the set  $\{(a_1, \dots, a_n) \in \mathbf{k}^n : f(a_1, \dots, a_n) = 0, \forall f \in \mathfrak{a}\}$ .

## 1.2 Associated prime ideals and Primary Decomposition

### 1.2.1 Associated prime ideals

**Definition 1.2.1** (Associated prime ideals). Let  $A$  be a noetherian ring and  $M$  an  $A$ -module. The *associated prime ideals* of  $M$  are the prime ideals  $\mathfrak{p}$  of form  $\text{Ann}(x)$  for some  $x \in M$ . The set of associated prime ideals of  $M$  is denoted by  $\text{Ass}(M)$ .

**Example 1.2.2.** Let  $A = \mathbf{k}[x, y]/(xy)$  and  $M = A$ . First we see that  $(x) = \text{Ann } y, (y) = \text{Ann } x \in \text{Ass } M$ . Then we check other prime ideals. For  $(x, y)$ , if  $xf = yf = 0$ , then  $f \in (x) \cap (y) = (0)$ . If  $(x - a) = \text{Ann } f$  for some  $f$ , note that  $y \in (x - a)$  for  $a \in \mathbf{k}^*$ , then  $f \in (x)$ . Hence  $f = 0$ . Therefore  $\text{Ass } M = \{(x), (y)\}$ .

**Example 1.2.3.** Let  $A = \mathbf{k}[x, y]/(x^2, xy)$  and  $M = A$ . The underlying space of  $\text{Spec } A$  is the  $y$ -axis since  $\sqrt{(x^2, xy)} = (x)$ . First note that  $(x) = \text{Ann } y, (x, y) = \text{Ann } x \in \text{Ass } M$ . For  $(x, y - a)$  with  $a \in \mathbf{k}^*$ , easily see that  $xf = (y - a)f = 0$  implies  $f = 0$  since  $A = \mathbf{k} \cdot x \oplus \mathbf{k}[y]$  as  $\mathbf{k}$ -vector space. Hence  $\text{Ass } M = \{(x), (x, y)\}$ .

**Lemma 1.2.4.** Let  $A$  be a noetherian ring and  $M$  an  $A$ -module. Then the maximal element of the set

$$\{\text{Ann } x : x \in M_{\mathfrak{p}}, x \neq 0\}$$

belongs to  $\text{Ass } M$ .



*Proof.* We just need to show that such  $\text{Ann } x$  is prime. Otherwise, there exist  $a, b \in A$  such that  $ab \in \text{Ann } x$  but  $a, b \notin \text{Ann } x$ . It follows that  $\text{Ann } x \subsetneq \text{Ann } ax$  since  $b \in \text{Ann } ax \setminus \text{Ann } x$ . This contradicts the maximality of  $\text{Ann } x$ .  $\square$

An element  $a \in A$  is called a zero divisor for  $M$  if  $M \rightarrow aM, m \mapsto am$  is not injective.

**Corollary 1.2.5.** Let  $A$  be a noetherian ring and  $M$  an  $A$ -module. Then

$$\{\text{zero divisors for } M\} = \bigcup_{\mathfrak{p} \in \text{Ass } M} \mathfrak{p}.$$

**Lemma 1.2.6.** Let  $A$  be a noetherian ring and  $M$  an  $A$ -module. Then  $\mathfrak{p} \in \text{Ass}_A M$  iff  $\mathfrak{p}A_{\mathfrak{p}} \in \text{Ass}_{A_{\mathfrak{p}}} M_{\mathfrak{p}}$ .

*Proof.* Suppose  $\mathfrak{p}A_{\mathfrak{p}} \in \text{Ass}_{A_{\mathfrak{p}}} M_{\mathfrak{p}}$ . Let  $\mathfrak{p}A_{\mathfrak{p}} = \text{Ann } y_0/c$  with  $y_0 \in M$  and  $c \in A \setminus \mathfrak{p}$ . For  $a \in \text{Ann } y_0$ ,  $ay_0 = 0$ . Then  $a/1 \in \mathfrak{p}A_{\mathfrak{p}}$ . It follows that  $a \in \mathfrak{p}$ . Hence  $\text{Ann } y_0 \subset \mathfrak{p}$ .

Inductively, if  $\text{Ann } y_n \subsetneq \mathfrak{p}$ , then there exists  $b_n \in A \setminus \mathfrak{p}$  such that  $y_{n+1} := b_n y_n$ ,  $\text{Ann } y_{n+1} \subset \mathfrak{p}$  and  $\text{Ann } y_n \subsetneq \text{Ann } y_{n+1}$ . To see this, choose  $a_n \in \mathfrak{p} \setminus \text{Ann } y_n$ . Then  $(a_n/1)y_n = 0$  since  $a_n/1 \in \mathfrak{p}A_{\mathfrak{p}}$ . By definition, there exist  $b_n \in A \setminus \mathfrak{p}$  such that  $a_n b_n y_n = 0$ . This process must terminate since  $A$  is noetherian. Thus  $\text{Ann } y_n = \mathfrak{p}$  for some  $n$ . Hence  $\mathfrak{p} \in \text{Ass}_A M$ .

Conversely, suppose  $\mathfrak{p} = \text{Ann } x \in \text{Ass } M$ . If  $(a/s)(x/1) = 0 \in M_{\mathfrak{p}}$ , there exist  $t \in A \setminus \mathfrak{p}$  such that  $tax = 0$ . It follows that  $ta \in \mathfrak{p}$  and then  $(a/s) \in \mathfrak{p}A_{\mathfrak{p}}$ . Hence  $\mathfrak{p}A_{\mathfrak{p}} \in \text{Ass}_{A_{\mathfrak{p}}} M_{\mathfrak{p}}$ .  $\square$

**Proposition 1.2.7.** We have  $\text{Ass } M \subset \text{Supp } M$ . Moreover, if  $\mathfrak{p} \in \text{Supp } M$  satisfies  $V(\mathfrak{p})$  is an irreducible component of  $\text{Supp } M$ , then  $\mathfrak{p} \in \text{Ass } M$ .

*Proof.* For any  $\mathfrak{p} = \text{Ann } x \in \text{Ass } M$ , we have  $A/\mathfrak{p} \cong A \cdot x \subset M$ . Tensoring with  $A_{\mathfrak{p}}$  gives  $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \hookrightarrow M_{\mathfrak{p}}$  since  $A_{\mathfrak{p}}$  is flat. Hence  $M_{\mathfrak{p}} \neq 0$  and  $\mathfrak{p} \in \text{Supp } M$ .

Now suppose  $\mathfrak{p} \in \text{Supp } M$  and  $V(\mathfrak{p})$  is an irreducible component of  $\text{Supp } M$ . First we show that  $\mathfrak{p} \in \text{Ass}_{A_{\mathfrak{p}}} M_{\mathfrak{p}}$ . Let  $x \in M_{\mathfrak{p}}$  such that  $\text{Ann } x$  is maximal in the set

$$\{\text{Ann } x : x \in M_{\mathfrak{p}}, x \neq 0\}.$$

Then we claim that  $\text{Ann } x = \mathfrak{p}A_{\mathfrak{p}}$ . First,  $\text{Ann } x$  is prime by Lemma 1.2.4. If  $\text{Ann } x \neq \mathfrak{p}$ , then  $V(\text{Ann } x) \supset V(\mathfrak{p})$ . This implies that  $\text{Ann } x \notin \text{Supp } M_{\mathfrak{p}}$  since  $\text{Supp } M_{\mathfrak{p}} = \text{Supp } M \cap \text{Spec } A_{\mathfrak{p}}$ . This is a contradiction. Thus  $\mathfrak{p}A_{\mathfrak{p}} \in \text{Ass}_{A_{\mathfrak{p}}} M_{\mathfrak{p}}$ . By Lemma 1.2.6, we have  $\mathfrak{p} \in \text{Ass } M$ .  $\square$

**Remark 1.2.8.** The existence of irreducible component is guaranteed by Zorn's Lemma.

**Definition 1.2.9.** A prime ideal  $\mathfrak{p} \in \text{Ass } M$  is called *embedded* if  $V(\mathfrak{p})$  is not an irreducible component of  $\text{Supp } M$ .

**Example 1.2.10.** For  $M = A = k[x, y]/(x^2, xy)$ , the origin  $(x, y)$  is an embedded point.

**Proposition 1.2.11.** If we have exact sequence  $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3$ , then  $\text{Ass } M_2 \subset \text{Ass } M_1 \cup \text{Ass } M_3$ .

*Proof.* Let  $\mathfrak{p} = \text{Ann } x \in \text{Ass } M_2 \setminus \text{Ass } M_1$ . Then the image  $[x]$  of  $x$  in  $M_3$  is not equal to 0. We have that  $\text{Ann } x \subset \text{Ann}[x]$ . If  $a \in \text{Ann}[x] \setminus \text{Ann } x$ , then  $ax \in M_1$ . Since  $\text{Ann } x \subsetneq \text{Ann } ax$ , there is  $b \in \text{Ann } ax \setminus \text{Ann } x$ . However, it implies  $ba \in \text{Ann } x$ , and then  $a \in \text{Ann } x$  since  $\text{Ann } x$  is prime,

which is a contradiction.  $\square$

**Corollary 1.2.12.** If  $M$  is finitely generated, then the set  $\text{Ass } M$  is finite.

*Proof.* For  $\mathfrak{p} = \text{Ann } x \in \text{Ass } M$ , we know that the submodule  $M_1$  generated by  $x$  is isomorphic to  $A/\mathfrak{p}$ . Inductively, we can choose  $M_n$  be the preimage of a submodule of  $M/M_{n-1}$  which is isomorphic to  $A/\mathfrak{q}$  for some  $\mathfrak{q} \in \text{Ass } M/M_{n-1}$ . We can take an ascending sequence  $0 = M_0 \subset M_1 \subset \cdots \subset M_n \subset \cdots$  such that  $M_i/M_{i-1} \cong A/\mathfrak{p}_i$  for some prime  $\mathfrak{p}_i$ . Since  $M$  is finitely generated, this is a finite sequence. Then the conclusion follows by Proposition 1.2.11.  $\square$

## 1.2.2 Primary decomposition

**Definition 1.2.13.** An  $A$ -module is called *co-primary* if  $\text{Ass } M$  has a single element. Let  $M$  be an  $A$ -module and  $N \subset M$  a submodule. Then  $N$  is called *primary* if  $M/N$  is co-primary. If  $\text{Ass } M/N = \{\mathfrak{p}\}$ , then  $N$  is called  $\mathfrak{p}$ -primary.

**Remark 1.2.14.** This definition coincide with primary ideals in the case  $M = A$ . Recall an ideal  $\mathfrak{q} \subset A$  is called *primary* if  $\forall ab \in \mathfrak{p}, a \notin \mathfrak{q}$  implies  $b^n \in \mathfrak{q}$  for some  $n$ .

Let  $\mathfrak{q}$  be a  $\mathfrak{q}$ -primary ideal. Since  $\text{Supp } A/\mathfrak{q} = \{\mathfrak{p}\}$ ,  $\mathfrak{p} \in \text{Ass } A/\mathfrak{q}$ . Suppose  $\text{Ann}[a] \in \text{Ass } A/\mathfrak{q}$ . Then  $\mathfrak{p} \subset \text{Ann}[a]$  since  $V(\mathfrak{p}) = \text{Supp } A/\mathfrak{q}$ . If  $b \in \text{Ann}[a]$ , then  $ab \in \mathfrak{q}$  and  $a \notin \mathfrak{q}$ . Hence  $b^n \in \mathfrak{q}$ , and then  $b \in \mathfrak{p}$ . This shows that  $\text{Ass } A/\mathfrak{q} = \{\mathfrak{p}\}$  and  $\mathfrak{q}$  is  $\mathfrak{p}$ -primary as an  $A$ -submodule.

Let  $\mathfrak{q} \subset A$  be a  $\mathfrak{p}$ -primary  $A$ -submodule. First we have  $\mathfrak{p} = \sqrt{\mathfrak{q}}$  since  $V(\mathfrak{p})$  is the unique irreducible component of  $\text{Supp } A/\mathfrak{q}$ . Suppose  $ab \in \mathfrak{q}$  and  $a \notin \mathfrak{q}$ . Then  $b \in \text{Ann}[a] \subset \mathfrak{p}$  since  $\mathfrak{p}$  is the unique maximal element in  $\{\text{Ann}[c] : c \in A \setminus \mathfrak{q}\}$ . This implies that  $b^n \in \mathfrak{q}$ .

**Definition 1.2.15.** Let  $A$  be a noetherian ring,  $M$  an  $A$ -module and  $N \subset M$  a submodule. A *minimal primary decomposition* of  $N$  in  $M$  is a finite set of primary submodules  $\{Q_i\}_{i=1}^n$  such that

$$N = \bigcap_{i=1}^n Q_i,$$

no  $Q_i$  can be omitted and  $\text{Ass } M/Q_i$  are pairwise distinct. For  $\text{Ass } M/Q_i = \{\mathfrak{p}\}$ ,  $Q_i$  is called belonging to  $\mathfrak{p}$ .

Indeed, if  $N \subset M$  admits a minimal primary decomposition  $N = \bigcap Q_i$  with  $Q_i$  belonging to  $\mathfrak{p}$ , then  $\text{Ass}(M/N) = \{\mathfrak{p}_i\}$ . For given  $i$ , consider  $N_i := \bigcap_{j \neq i} Q_j$ , then  $N_i/N \cong (N_i + Q_i)/Q_i$ . Since  $N_i \neq N$ ,  $\text{Ass } N_i/N \neq \emptyset$ . On the other hand,  $\text{Ass } N_i/N \subset \text{Ass } M/Q_i = \{\mathfrak{p}\}$ . It follows that  $\text{Ass } N_i/N = \{\mathfrak{p}_i\}$ , whence  $\mathfrak{p}_i \in \text{Ass } M/N$ . Conversely, we have an injection  $M/N \hookrightarrow \bigoplus M/Q_i$ , so  $\text{Ass } M/N \subset \bigcup \text{Ass } M/Q_i$ . Due to this, if  $Q_i$  belongs to  $\mathfrak{p}$ , we also say that  $Q_i$  is the  $\mathfrak{p}$ -component of  $N$ .

**Proposition 1.2.16.** Suppose  $N \subset M$  has a minimal primary decomposition. If  $\mathfrak{p} \in \text{Ass } M/N$  is not embedded, then the  $\mathfrak{p}$  component of  $N$  is unique. Explicitly, we have  $Q = \nu^{-1}(N_{\mathfrak{p}})$ , where  $\nu : M \rightarrow M_{\mathfrak{p}}$ .

*Proof.* First we show that  $Q = \nu^{-1}(Q_{\mathfrak{p}})$ . Clearly  $Q \subset \nu^{-1}(Q_{\mathfrak{p}})$ . Suppose  $x \in \nu^{-1}(Q_{\mathfrak{p}})$ . Then there exists  $s \in A \setminus \mathfrak{p}$  such that  $sx \in Q$ . That is,  $[sx] = 0 \in M/Q$ . If  $[x] \neq 0$ , we have  $s \in \text{Ann}[x] \subset \mathfrak{p}$ . This contradiction enforces  $Q = \nu^{-1}(Q_{\mathfrak{p}})$ .

Then we show that  $N_{\mathfrak{p}} = Q_{\mathfrak{p}}$ . Just need to show that for  $\mathfrak{p}' \neq \mathfrak{p}$  and the  $\mathfrak{p}'$  component  $Q'$  of  $N$ ,  $Q'_{\mathfrak{p}} = M_{\mathfrak{p}}$ . Since  $\mathfrak{p}$  is not embedded,  $\mathfrak{p}' \not\subset \mathfrak{p}$ . Then  $\mathfrak{p} \notin V(\mathfrak{p}) = \text{Supp } M/Q'$ . So  $M_{\mathfrak{p}}/Q'_{\mathfrak{p}} = 0$ .  $\square$

**Example 1.2.17.** If  $\mathfrak{p}$  is embedded, then its components may not be unique. For example, let  $M = A = \mathbb{k}[x, y]/(x^2, xy)$ . Then for every  $n \in \mathbb{Z}_{\geq 1}$ ,  $(x) \cap (x^2, xy, y^n)$  is a minimal primary decomposition of  $(0) \subset M$ .

Let  $A$  be a noetherian ring and  $\mathfrak{p} \subset A$  a prime ideal. We consider the  $\mathfrak{p}$  component of  $\mathfrak{p}^n$ , which is called  $n$ -th symbolic power of  $\mathfrak{p}$ , denoted by  $\mathfrak{p}^{(n)}$ . We have  $\mathfrak{p}^{(n)} = \mathfrak{p}^n A_{\mathfrak{p}} \cap A$ . In general,  $\mathfrak{p}^{(n)}$  is not equal to  $\mathfrak{p}^n$ ; see below example.

**Example 1.2.18.** Let  $A = \mathbb{k}[x, y, z, w]/(y^2 - zx^2, yz - xw)$  and  $\mathfrak{p} = (y, z, w)$ . We have  $z = y^2/x^2, w = yz/x \in \mathfrak{p}^2 A_{\mathfrak{p}}$ , whence  $\mathfrak{p}^2 A_{\mathfrak{p}} = (z, w) \neq \mathfrak{p}^2$ .

**Theorem 1.2.19.** Let  $A$  be a noetherian ring and  $M$  an  $A$ -module. Then for every  $\mathfrak{p} \in \text{Ass } M$ , there is a  $\mathfrak{p}$ -primary submodule  $Q(\mathfrak{p})$  such that

$$(0) = \bigcap_{\mathfrak{p} \in \text{Ass } M} Q(\mathfrak{p}).$$

*Proof.* Consider the set

$$\mathcal{N} := \{N \subset M : \mathfrak{p} \notin \text{Ass } N\}.$$

Note that  $\text{Ass } \bigcup N_i = \bigcup \text{Ass } N_i$  by definition of associated prime ideals. Then it is easy to check that  $\mathcal{N}$  satisfies the conditions of Zorn's Lemma. Hence  $\mathcal{N}$  has a maximal element  $Q(\mathfrak{p})$ . We claim that  $Q(\mathfrak{p})$  is  $\mathfrak{p}$ -primary. If there is  $\mathfrak{p}' \neq \mathfrak{p} \in \text{Ass } M/Q(\mathfrak{p})$ , then there is a submodule  $N' \cong A/\mathfrak{p}'$ . Let  $N''$  be the preimage of  $N'$  in  $M$ . We have  $Q(\mathfrak{p}) \subsetneq N''$  and  $N'' \in \mathcal{N}$ . This is a contradiction. By the fact  $\text{Ass } \bigcap N_i = \bigcap \text{Ass } N_i$ , we get the conclusion.  $\square$

**Corollary 1.2.20.** Let  $A$  be a noetherian ring and  $M$  a finite  $A$ -module. Then every submodule of  $M$  has a minimal primary decomposition.

## 1.3 Rings of Lower Dimension

**Definition 1.3.1.** Let  $A$  be a noetherian ring. The *height of a prime ideal*  $\mathfrak{p}$  in  $A$  is defined as the maximum length of chains of prime ideals contained in  $\mathfrak{p}$ , that is,

$$\text{ht}(\mathfrak{p}) := \sup\{n \mid \exists \text{ a chain of prime ideals } \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{p}\}.$$

The *Krull dimension* of  $A$  is defined as

$$\dim A := \max_{\mathfrak{p} \in \text{Spec } A} \text{ht}(\mathfrak{p}).$$

**Example 1.3.2.** Let  $A$  be a PID. For every two non-zero prime ideals  $\mathfrak{p}_1$  and  $\mathfrak{p}_2$ , if  $\mathfrak{p}_1 = t_1 A \subset \mathfrak{p}_2 = t_2 A$ , then  $t_2 \mid t_1$  and hence  $\mathfrak{p}_1 = \mathfrak{p}_2$ . It follows that  $\dim A = 1$ . Consequently, the ring of integers  $\mathbb{Z}$  and the polynomial ring  $\mathbb{k}[T]$  in one variable over a field have Krull dimension 1.

### 1.3.1 Artinian Rings and Length of Modules

**Definition 1.3.3.** Let  $A$  be a ring and  $M$  an  $A$  module. A *simple module filtration* of  $M$  is a filtration

$$M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_n = 0$$

such that  $M_i/M_{i-1}$  is a simple module, i.e. it has no submodule except  $0$  and itself. If  $M$  has a simple module filtration as above, we define the *length* of  $M$  as  $n$  and say that  $M$  has *finite length*.

The following proposition guarantees the length is well-defined.

**Proposition 1.3.4.** Suppose  $M$  has a simple module filtration  $M = M_{0,0} \supsetneq M_{1,0} \supsetneq \cdots \supsetneq M_{n,0} = 0$ . Then for any other filtration  $M = M_{0,0} \supset M_{0,1} \supset \cdots \supset M_{0,m} = 0$  with  $m > n$ , there exist  $k < m$  such that  $M_{0,k} = M_{0,k+1}$ .

*Proof.* We claim that there are at least  $0 \leq k_1 < \cdots < k_{m-n} < m$  satisfies that  $M_{0,k_i} = M_{0,k_i+1}$ . Let  $M_{i,j} := M_{i,0} \cap M_{0,j}$ . Inductively on  $n$ , we can assume that there exist  $k_1, \dots, k_{n-m+1}$  such that  $M_{1,k} = M_{1,k+1}$ . Consider the sequence

$$M_{0,0}/M_{1,0} \supset (M_{0,1} + M_{1,0})/M_{1,0} \supset \cdots \supset (M_{0,m} + M_{1,0})/M_{1,0} = 0$$

in  $M_{0,0}/M_{1,0}$ . Since  $M_{0,0}/M_{1,0}$  is simple, there is at most one  $k_i$  with  $M_{0,k_i} + M_{1,0} \neq M_{0,k_i+1} + M_{1,0}$ . And note that if  $M_{0,k_i} + M_{1,0} = M_{0,k_i+1} + M_{1,0}$  and  $M_{0,k_i} \cap M_{1,0} = M_{0,k_i+1} \cap M_{1,0}$ , then  $M_{0,k_i} = M_{0,k_i+1}$  by the Five Lemma.  $\square$

**Example 1.3.5.** Let  $A$  be a ring and  $\mathfrak{m} \in \text{mSpec } A$ . Then  $A/\mathfrak{m}$  is a simple module. Yang: To be completed.

**Proposition 1.3.6.** Let  $A$  be a ring and  $M$  an  $A$ -module. Then  $M$  is of finite length iff it satisfies both a.c.c and d.c.c.

*Proof.* Note that if  $M$  has either a strictly ascending chain or a strictly descending chain,  $M$  is of infinite length. Conversely, d.c.c guarantee  $M$  has a simple submodule and a.c.c guarantee the sequence terminates.  $\square$

**Proposition 1.3.7.** The length  $l(-)$  is an additive function for modules of finite length. That is, if we have an exact sequence  $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$  with  $M_i$  of finite length, then  $l(M_2) = l(M_1) + l(M_3)$ .

*Proof.* The simple module filtrations of  $M_1$  and  $M_3$  will give a simple module filtration of  $M_2$ .  $\square$

**Proposition 1.3.8.** Let  $(A, \mathfrak{m})$  be a local ring. Then  $A$  is artinian iff  $\mathfrak{m}^n = 0$  for some  $n \geq 0$ .

*Proof.* Suppose  $A$  is artinian. Then the sequence  $\mathfrak{m} \supset \mathfrak{m}^2 \supset \mathfrak{m}^3 \supset \cdots$  is stable. It follows that  $\mathfrak{m}^n = \mathfrak{m}^{n+1}$  for some  $n$ . By the Nakayama's Lemma 1.1.14,  $\mathfrak{m}^n = 0$ .

Conversely, we have

$$\mathfrak{m} \subset \mathfrak{N} \subset \bigcap_{\text{minimal prime ideal}} \mathfrak{p},$$

whence  $\mathfrak{m}$  is minimal.  $\square$

**Proposition 1.3.9.** Let  $A$  be a ring. Then  $A$  is artinian iff  $A$  is of finite length.

*Proof.* First we show that  $A$  has only finite maximal ideal. Otherwise, consider the set  $\{\mathfrak{m}_1 \cap \mathfrak{m}_2 \cap \cdots \cap \mathfrak{m}_k\}$ . It has a minimal element  $\mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n$  and for any maximal ideal  $\mathfrak{m}$ ,  $\mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n \subset \mathfrak{m}$ . It follows that  $\mathfrak{m} = \mathfrak{m}_i$  for some  $i$ . Let  $\mathfrak{M} = \mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n$  be the Jacobi radical of  $A$ . Consider the sequence  $\mathfrak{M} \supset \mathfrak{M}^2 \supset \cdots$  and by Nakayama's Lemma, we have  $\mathfrak{M}^k = 0$  for some  $k$ . Consider the filtration

$$A \supset \mathfrak{m}_1 \supset \cdots \supset \mathfrak{m}_1^k \supset \mathfrak{m}_1^k \mathfrak{m}_2 \supset \cdots \supset \mathfrak{m}_1^k \cdots \mathfrak{m}_n^k = (0).$$

We have  $\mathfrak{m}_1^k \cdots \mathfrak{m}_i^j / \mathfrak{m}_1^k \cdots \mathfrak{m}_i^{j+1}$  is an  $A/\mathfrak{m}_i$ -vector space. It is artinian and then of finite length. Hence  $A$  is of finite length.  $\square$

**Theorem 1.3.10.** Let  $A$  be a ring. Then  $A$  is artinian iff  $A$  is noetherian and of dimension 0.

*Proof.* Suppose  $A$  is artinian. Then  $A$  is noetherian by Proposition 1.3.9. Let  $\mathfrak{p} \in \text{Spec } A$ . Then  $A/\mathfrak{p}$  is an artinian integral domain. If there is  $a \in A/\mathfrak{p}$  is not invertible, consider  $(a) \supset (a^2) \supset \cdots$ , we see  $a = 0$ . Hence  $\mathfrak{p}$  is maximal and  $\dim A = 0$ .

Suppose that  $A$  is noetherian and of dimension 0. Then every maximal ideal is minimal. In particular,  $A$  has only finite maximal ideal  $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ . Let  $\mathfrak{q}_i$  be the  $\mathfrak{p}_i$ -component of  $(0)$ . Then we have  $A \hookrightarrow \bigoplus_i A/\mathfrak{q}_i$ . We just need to show that  $A/\mathfrak{q}_i$  is of finite length as  $A$ -module. If  $\mathfrak{q}_i \subset \mathfrak{p}_j$ , take radical we get  $\mathfrak{p}_i \subset \mathfrak{p}_j$  and hence  $i = j$ . So  $A/\mathfrak{q}_i$  is a local ring with maximal ideal  $\mathfrak{p}_i A/\mathfrak{q}_i$ . Then every element in  $\mathfrak{p}_i A/\mathfrak{q}_i$  is nilpotent. Since  $\mathfrak{p}_i$  is finitely generated,  $(\mathfrak{p}_i A/\mathfrak{q}_i)^k = 0$  for some  $k$ . Then  $A/\mathfrak{q}_i$  is artinian and then of finite length as  $A/\mathfrak{q}_i$ -module. Then the conclusion follows.  $\square$

### 1.3.2 DVR and Dedekind Domain

**Definition 1.3.11.** A *valuation* on a field  $K$  is a function  $v : K^\times \rightarrow \Gamma$ , where  $\Gamma$  is a totally ordered abelian group, such that for all  $x, y \in K^\times$ :

- (i)  $v(xy) = v(x) + v(y)$ .
- (ii)  $v(x + y) \geq \min(v(x), v(y))$  if  $x + y \neq 0$ .

We extend  $v$  to  $K$  by setting  $v(0) = +\infty$ , where  $+\infty$  is an element greater than all elements of  $\Gamma$ . If  $\Gamma$  is discrete with respect to the order topology, we say that  $v$  is a *discrete valuation*.

- Example 1.3.12.** (a) Let  $\mathbf{K} = \mathbb{Q}$  and  $p$  be a prime number. Let  $v : \mathbb{Q}^\times \rightarrow \mathbb{Z}$  be defined by  $v(a/b) = n$  if  $a/b = p^n(c/d)$  with  $c, d$  coprime to  $p$ . Then  $v$  is a discrete valuation on  $\mathbb{Q}$ .
- (b) Let  $\mathbf{K} = \mathbf{k}(T)$  be the field of rational functions over a field  $\mathbf{k}$ . For  $f = x^n g \in \mathbf{k}(t)$  with  $g(0) \in \mathbf{k}^\times$ , let  $v(f) = n$ .
- (c) Let  $\mathbf{K} = \mathbb{C}_p$  the  $p$ -adic complex numbers. For  $x \in \mathbb{C}_p^\times$ , let  $v(x) = -\log_p |x|_p$ . Then  $v$  is a valuation on  $\mathbb{C}_p$  which is not discrete.

**Definition 1.3.13** (Discrete Valuation Ring). Let  $V$  be a domain with field of fractions  $\mathbf{K}$ . We say that  $V$  is a *discrete valuation ring (DVR)* if there exists a discrete valuation  $v$  on  $\mathbf{K}$  such that  $V = \{x \in \mathbf{K} \mid v(x) \geq 0\}$ .

Yang: To be completed

## 1.4 Dimension and Depth

There are three numbers measuring the “size” of a local ring  $(A, \mathfrak{m})$ :

- $\dim A$ : the Krull dimension of  $A$ .
- $\text{depth } A$ : the depth of  $A$ .
- $\dim_{\kappa(\mathfrak{m})} T_{A, \mathfrak{m}}$ : the dimension of Zariski tangent space  $T_{A, \mathfrak{m}} := (\mathfrak{m}/\mathfrak{m}^2)^\vee$  as a  $\kappa(\mathfrak{m})$ -vector space.

Somehow the Krull dimension is “homological” and the depth is “cohomological”.

**Definition 1.4.1.** Let  $A$  be a noetherian ring,  $I \subset A$  an ideal and  $M$  a finitely generated  $A$ -module. A sequence  $t_1, \dots, t_n \in I$  is called an  *$M$ -regular sequence in  $I$*  if  $t_i$  is not a zero divisor on  $M/(t_1, \dots, t_{i-1})M$  for all  $i$ .

**Example 1.4.2.** Let  $A = \mathbf{k}[x, y]/(x^2, xy)$  and  $I = (x, y)$ . Then  $\text{depth}_I A = 0$ .

**Definition 1.4.3.** Let  $A$  be a noetherian ring. For every  $\mathfrak{p} \in \text{Spec } A$ ,  $\mathfrak{p}/\mathfrak{p}^2$  is a vector space over  $\kappa(\mathfrak{p})$ . The *Zariski’s tangent space*  $T_{A, \mathfrak{p}}$  of  $A$  at  $\mathfrak{p}$  is defined as  $(\mathfrak{p}/\mathfrak{p}^2)^\vee$ , the dual  $\kappa(\mathfrak{p})$ -vector space of  $\mathfrak{p}/\mathfrak{p}^2$ .

### 1.4.1 Krull’s Principal Ideal Theorem

**Theorem 1.4.4** (Krull’s Principal Ideal Theorem). Let  $A$  be a noetherian ring. Suppose  $f \in A$  is not a unit. Let  $\mathfrak{p}$  be a minimal prime ideal among those containing  $f$ . Then  $\text{ht}(\mathfrak{p}) \leq 1$ .

*Proof.* By replacing  $A$  by  $A_{\mathfrak{p}}$ , we may assume  $A$  is local with maximal ideal  $\mathfrak{p}$ . Note that  $A/(f)$  is artinian since it has only one prime ideal  $\mathfrak{p}/(f)$ .

Let  $\mathfrak{q} \subsetneq \mathfrak{p}$ . Consider the sequence  $\mathfrak{q}^{(1)} \supset \mathfrak{q}^{(2)} \supset \dots$ , its image in  $A/(f)$  is stationary. Then there exists  $n \in \mathbb{Z}_{\geq 0}$  such that  $\mathfrak{q}^{(n)} + (f) = \mathfrak{q}^{(n+1)} + (f)$ . For  $x \in \mathfrak{q}^{(n)}$ , we may write  $x = y + af$  for  $y \in \mathfrak{q}^{(n+1)}$ . Then  $af \in \mathfrak{q}^{(n)}$ . Since  $\mathfrak{q}^{(n)}$  is  $\mathfrak{q}$ -primary and  $f \notin \mathfrak{q}$ ,  $a \in \mathfrak{q}^{(n)}$ . Then we get  $\mathfrak{q}^{(n)} = \mathfrak{q}^{(n+1)} + f\mathfrak{q}^{(n)}$ . That is,  $\mathfrak{q}^{(n)}/\mathfrak{q}^{(n+1)} = f\mathfrak{q}^{(n)}/\mathfrak{q}^{(n+1)}$ . Note that  $f \in \mathfrak{p}$ , by Nakayama’s Lemma,  $\mathfrak{q}^{(n)} = \mathfrak{q}^{(n+1)}$ . That is,  $\mathfrak{q}^n A_{\mathfrak{q}} = \mathfrak{q}^{n+1} A_{\mathfrak{q}}$ . By Nakayama’s Lemma again,  $\mathfrak{q}^n A_{\mathfrak{q}} = 0$ . It follows that  $\mathfrak{q} A_{\mathfrak{q}}$  is minimal, whence  $A_{\mathfrak{q}}$  is artinian. Therefore,  $\mathfrak{q}$  is minimal in  $A$ .  $\square$

**Corollary 1.4.5.** Let  $A$  be a noetherian local ring. Suppose  $f \in A$  is not a unit. Then  $\dim A/(f) \geq \dim A - 1$ . If  $f$  is not contained in a minimal prime ideal, the equality holds.

*Proof.* Let  $\mathfrak{p}_0 \subsetneq \dots \subsetneq \mathfrak{p}_n$  be a sequence of prime ideals. By assumption,  $f \in \mathfrak{p}_n$ . If  $f \in \mathfrak{p}_0$ , we get a sequence of prime ideals in  $A/(f)$  of length  $n$ . Now we suppose  $f \notin \mathfrak{p}_0$ . Then there exists  $k \geq 0$  such that  $f \in \mathfrak{p}_{k+1} \setminus \mathfrak{p}_k$ .

Choose  $\mathfrak{q}$  be a minimal prime ideal among those containing  $(\mathfrak{p}_{k-1}, f)$  and contained in  $\mathfrak{p}_{k+1}$ . Then by Krull’s Principal Ideal Theorem 1.4.4,  $\mathfrak{q}_k \subsetneq \mathfrak{p}_{k+1}$ . Replace  $\mathfrak{p}_k$  by  $\mathfrak{q}_k$ , we have  $f \in \mathfrak{q}_k \setminus \mathfrak{p}_{k-1}$

Repeat this process, we get a sequence  $\mathfrak{p}'_0 \subsetneq \cdots \subsetneq \mathfrak{p}'_n$  such that  $f \in \mathfrak{p}'_1$ . This gives a sequence  $\mathfrak{p}'_1 \subsetneq \cdots \subsetneq \mathfrak{p}'_n$  in  $A/(f)$ . Hence we get  $\dim A/(f) \geq \dim A - 1$ .

Since  $f$  is not contained in minimal prime ideal, preimage of a minimal prime ideal in  $A/(f)$  has height 1. Hence a sequence of prime ideals in  $A/fA$  can be extended by a minimal prime ideal in  $A$ . It follows that  $\dim A/(f) + 1 \leq \dim A$ .  $\square$

**Proposition 1.4.6.** Let  $(A, \mathfrak{m})$  be a local noetherian ring with residue field  $\mathbf{k}$ . Then the following inequalities hold:

$$\text{depth } A \leq \dim A \leq \dim_{\mathbf{k}} T_{A, \mathfrak{m}}.$$

*Proof.* The first inequality is a direct corollary of Corollary 1.4.5.

Let  $t_1, \dots, t_n$  be a  $\kappa(\mathfrak{m})$ -basis of  $\mathfrak{m}/\mathfrak{m}^2$ . Then we have  $\mathfrak{m}/(t_1, \dots, t_n) + \mathfrak{m}^2 = 0$ , whence  $\mathfrak{m}/(t_1, \dots, t_n) = \mathfrak{m}(\mathfrak{m}/(t_1, \dots, t_n))$ . It follows that  $\mathfrak{m} = (t_1, \dots, t_n)$  by Nakayama's Lemma. By Corollary 1.4.5,

$$n + \dim A/(t_1, \dots, t_n) \geq n - 1 + \dim A/(t_1, \dots, t_{n-1}) \geq \cdots \geq 1 + \dim A/(t_1) \geq \dim A.$$

We conclude the result.  $\square$

**Definition 1.4.7.** Let  $X$  be a locally noetherian scheme and  $k \in \mathbb{Z}_{\geq 0}$ . We say that  $X$  *verifies property  $(R_k)$*  or *is regular in codimension  $k$*  if  $\forall \xi \in X$  with  $\text{codim } Z_\xi \leq k$ ,

$$\dim_{\kappa(\xi)} T_{X, \xi} = \dim \mathcal{O}_{X, \xi}.$$

We say that  $X$  *verifies property  $(S_k)$*  if  $\forall \xi \in X$  with  $\text{depth } \mathcal{O}_{X, \xi} < k$ ,

$$\text{depth } \mathcal{O}_{X, \xi} = \dim \mathcal{O}_{X, \xi}.$$

**Example 1.4.8.** Let  $A$  be a noetherian ring. Then  $A$  verifies  $(S_1)$  iff  $A$  has no embedded point.

Suppose  $A$  verifies  $(S_1)$ . If  $\mathfrak{p} \in \text{Ass } A$ , every element in  $\mathfrak{p}$  is a zero divisor. Then  $\text{depth } A_{\mathfrak{p}} = 0$ . It follows that  $\dim A_{\mathfrak{p}} = 0$  and then  $\mathfrak{p}$  is minimal.

Suppose  $A$  has no embedded point. Let  $\mathfrak{p} \in \text{Spec } A$  with  $\text{depth } A_{\mathfrak{p}} = 0$ . This means every element in  $\mathfrak{p}A_{\mathfrak{p}}$  is a zero divisor. Then

$$\mathfrak{p} \subset \{\text{zero divisors in } A\} = \bigcup_{\text{minimal prime ideals}} \mathfrak{q}.$$

By Proposition 1.1.2,  $\mathfrak{p} = \mathfrak{q}$  for some minimal  $\mathfrak{q}$ , whence  $\dim A_{\mathfrak{p}} = 0$ .

**Example 1.4.9.** Let  $A$  be a noetherian ring. Then  $A$  is reduced iff it verifies  $(R_0)$  and  $(S_1)$ .

Suppose  $A$  is reduced. Let  $\mathfrak{p}_1, \dots, \mathfrak{p}_n$  be all minimal prime ideals of  $A$ . We have  $\bigcap \mathfrak{p}_i = \mathfrak{N} = (0)$ , where  $\mathfrak{N}$  is the nilradical of  $A$ . Hence  $A$  has no embedded point. Since  $A_{\mathfrak{p}}$  is artinian, local and reduced,  $A_{\mathfrak{p}}$  is a field and hence regular.

Conversely, let  $\text{Ass } A$  be equal to  $\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$ . Then every  $\mathfrak{p}_i$  is minimal by  $(S_1)$ . Let  $f$  be in  $\mathfrak{N}$ . Then the image of  $f$  in  $A_{\mathfrak{p}_i}$  is 0 since by  $(R_0)$ ,  $A_{\mathfrak{p}_i}$  is a field. It follows that  $f \in \mathfrak{q}_i$ , where  $\mathfrak{q}_i$  is the  $\mathfrak{p}_i$  component of  $(0)$  in  $A$ . Hence  $f \in \bigcap \mathfrak{q}_i = (0)$ . That is,  $A$  is reduced.



### 1.4.2 Cohen-Macaulay rings

**Definition 1.4.10** (Cohen-Macaulay). A noetherian local ring  $(A, \mathfrak{m})$  is called *Cohen-Macaulay* if  $\dim A = \text{depth } A$ . A noetherian ring  $A$  is called *Cohen-Macaulay* if for every prime ideal  $\mathfrak{p} \in \text{Spec } A$ , the localization  $A_{\mathfrak{p}}$  is Cohen-Macaulay. This is equivalent to that  $A$  verifies  $(S_k)$  for all  $k \geq 0$ .

**Definition 1.4.11.** Let  $(A, \mathfrak{m})$  be a noetherian local ring of dimension  $d$ . A sequence  $t_1, \dots, t_d \in \mathfrak{m}$  is called a *system of parameters* if **Yang: To be completed.**

**Example 1.4.12** (Non Cohen-Macaulay rings). **Yang: To be completed.**

**Corollary 1.4.13.** Let  $A$  be a noetherian ring,  $M$  a finite  $A$ -module and  $a \in A$  an  $M$ -regular element. Then  $\text{depth } M = \text{depth } M/aM + 1$ .

**Corollary 1.4.14.** Let  $A$  be a noetherian ring  $a \in A$  a nonzero divisor. Then  $A$  verifies  $(S_d)$  iff  $A/aA$  verifies  $(S_{d-1})$ .

**Definition 1.4.15.** An ideal  $I$  of a noetherian ring  $A$  is called *unmixed* if

$$\text{ht}(I) = \text{ht}(\mathfrak{p}), \quad \forall \mathfrak{p} \in \text{Ass}(A/I).$$

Here  $\text{ht}(I)$  is defined as

$$\text{ht}(I) := \inf\{\text{ht}(\mathfrak{p}) : I \subset \mathfrak{p}\}.$$

We say that *the unmixedness theorem holds for a noetherian ring  $A$*  if any ideal  $I \subset A$  generated by  $\text{ht}(I)$  elements is unmixed. We say that *the unmixedness theorem holds for a locally noetherian scheme  $X$*  if  $\mathcal{O}_{X, \xi}$  is unmixed for any point  $\xi \in X$ .

**Proposition 1.4.16.** Let  $X$  be a locally noetherian scheme. Then the unmixedness theorem holds for  $X$  if and only if  $X$  is Cohen-Macaulay.

*Proof.* We can assume that  $X = \text{Spec } A$  is affine.

Suppose  $X$  is Cohen-Macaulay. Let  $I \subset A$  be an ideal generated by  $a_1, \dots, a_r$  with  $r = \text{ht}(I)$ . We claim that  $a_1, \dots, a_r$  is an  $A$ -regular sequence. If so, we get that the unmixedness theorem holds for  $A$  by applying Example 1.4.8 on  $A/I$ . Since  $\text{ht}(a_1, \dots, a_{r-1}) \leq r - 1$  by Krull's Principal Ideal Theorem 1.4.4 and  $\text{ht}(a_1, \dots, a_r) = r \leq \text{ht}(a_1, \dots, a_{r-1}) + 1$ , we have  $\text{ht}(a_1, \dots, a_{r-1}) = r - 1$ . By induction on  $r$ , we can assume that  $a_1, \dots, a_{r-1}$  is an  $A$ -regular sequence. Hence any prime ideal  $\mathfrak{p} \in \text{Ass } A/(a_1, \dots, a_{r-1})$  has height  $r - 1$ . Now suppose  $a_r$  is a zero divisor in  $A/(a_1, \dots, a_{r-1})$ . Then there exists a prime ideal  $\mathfrak{p} \in \text{Ass } A/(a_1, \dots, a_{r-1})$  such that  $a_r \in \mathfrak{p}$ . Then  $I \subset \mathfrak{p}$  and  $\text{ht}(I) \leq r - 1$ . This contradicts that  $\text{ht}(I) = r$ .

Suppose the unmixedness theorem holds for  $A$ . Let  $\mathfrak{p} \in \text{Spec } A$  be a prime ideal with  $\text{ht}(\mathfrak{p}) = r$ . Then  $\mathfrak{p} \in \text{Ass } A$  if and only if  $\text{ht}(\mathfrak{p}) = 0$ . If  $r > 0$ , there is a nonzero divisor  $a \in \mathfrak{p}$ . By Krull's Principal Ideal Theorem 1.4.4,  $\text{ht}(\mathfrak{p}A/aA) = r - 1$ . Inductively, we can find a regular sequence  $a_1, \dots, a_r$  in  $\mathfrak{p}$ . Then  $\text{depth } A_{\mathfrak{p}} = r$ .  $\square$



**Theorem 1.4.17.** Let  $A$  be a noetherian ring of dimension  $d$ . The following are equivalent: **Yang:**  
To be completed.

*Proof.* **Yang:** To be completed. □

### 1.4.3 Regular rings

**Definition 1.4.18.** A noetherian ring  $A$  is said to be *regular* at  $\mathfrak{p} \in \operatorname{Spec} A$  if we have

$$\dim_{\kappa(\mathfrak{p})} T_{A,\mathfrak{p}} = \dim A_{\mathfrak{p}},$$

where  $\dim A_{\mathfrak{p}}$  is the Krull dimension of the local ring  $A_{\mathfrak{p}}$ .

A noetherian ring  $A$  is said to be *regular* if it is regular at every prime ideal  $\mathfrak{p} \in \operatorname{Spec} A$ . This is equivalent to the condition that  $A$  verifies  $(R_k)$  for all  $k \geq 0$ .

**Remark 1.4.19.** A noetherian ring  $A$  is regular if and only if it is regular at every maximal ideal  $\mathfrak{m} \in \operatorname{mSpec} A$ . The proof uses homological tools; see Theorem 2.3.17 and Corollary 2.3.18.

**Definition 1.4.20.** Let  $A$  be a noetherian ring that is regular at  $\mathfrak{p} \in \operatorname{Spec} A$ . A sequence  $t_1, \dots, t_n \in \mathfrak{p}$  is called a *regular system of parameters* at  $\mathfrak{p}$  if their images form a basis of the  $\kappa(\mathfrak{p})$ -vector space  $\mathfrak{p}/\mathfrak{p}^2$ .

**Proposition 1.4.21.** Let  $(A, \mathfrak{m})$  be a noetherian local ring that is regular at  $\mathfrak{m}$ . Let  $t_1, \dots, t_n$  be a regular system of parameters at  $\mathfrak{m}$ ,  $\mathfrak{p}_i = (t_1, \dots, t_i)$  and  $\mathfrak{p}_0 = (0)$ . Then  $\mathfrak{p}_i$  is a prime ideal of height  $i$ , and  $A/\mathfrak{p}_i$  is a regular local ring for all  $i$ . In particular, regular local ring is integral, and the regular system of parameters  $t_1, \dots, t_n$  is a regular sequence in  $A$ .

*Proof.* By the Krull's Principal Ideal Theorem 1.4.4, we have

$$n - 1 = \dim A - 1 \leq \dim A/(t_1) \leq \dim_{\kappa(\mathfrak{m}/(t_1))} T_{A/(t_1), \mathfrak{m}/(t_1)} \leq n - 1.$$

Hence  $\dim A/(t_1) = n - 1$  and  $\operatorname{ht}(t_1) = 1$ . Since  $t_2, \dots, t_n$  generate  $\mathfrak{m}/(t_1)$ , we have that  $A/(t_1)$  is regular at  $\mathfrak{m}/(t_1)$  and the images of  $t_2, \dots, t_n$  form a regular system of parameters.

For integrality, we induct on the dimension of  $A$ . If  $\dim A = 0$ , then  $A$  is a field and hence integral. Suppose  $\dim A > 0$ , let  $\mathfrak{q}$  be a minimal prime ideal of  $A$ . Then  $t_1 \notin \mathfrak{q}$ . We have

$$n - 1 = \dim A - 1 \leq \dim A/(\mathfrak{q} + t_1 A) \leq \dim_{\kappa(\mathfrak{q}/(t_1))} T_{A/(\mathfrak{q} + t_1 A), \mathfrak{q}/(t_1)} \leq n - 1.$$

By similar arguments, we have  $A/(\mathfrak{q} + t_1 A)$  is regular at  $\mathfrak{m}/(\mathfrak{q} + t_1 A)$ . By induction hypothesis, both of  $A/t_1 A$  and  $A/(\mathfrak{q} + t_1 A)$  are integral and of dimension  $n - 1$ . Hence  $t_1 A = t_1 A + \mathfrak{q}$ , i.e.  $\mathfrak{q} \subset t_1 A$ . For every  $a = bt_1 \in \mathfrak{q}$ , we have  $b \in \mathfrak{q}$  since  $t_1 \notin \mathfrak{q}$ . Then  $\mathfrak{q} \subset t_1 \mathfrak{q} \subset \mathfrak{m} \mathfrak{q}$ . By Nakayama's Lemma,  $\mathfrak{q} = 0$ , whence  $A$  is integral. □

**Corollary 1.4.22.** A regular noetherian ring is Cohen-Macaulay.

**Corollary 1.4.23.** A regular noetherian ring is normal.

**Remark 1.4.24.** Indeed we can show a stronger result: a noetherian regular local ring is a UFD; see Yang: ref.

## 1.5 Finite Algebra and Normality

Let  $R$  be a ring and  $A$  be an  $R$ -algebra. We say that  $A$  is *of finite type* over  $R$  if there exists a surjective  $R$ -algebra homomorphism  $R[T_1, \dots, T_n] \rightarrow A$  for some  $n \geq 0$ . We say that  $A$  is *finite* over  $R$  if it is finite as an  $R$ -module.

### 1.5.1 Finite algebra

Let  $A$  be a ring and  $B$  a finite  $A$ -algebra.

**Example 1.5.1.** Let  $K$  be a number field. Then  $O_K$  is a finite  $\mathbb{Z}$ -algebra. Yang: To be completed.

**Lemma 1.5.2.** Let  $A \subset B$  be noetherian rings such that  $B$  is finite over  $A$ . Then the induced morphism  $\text{Spec } B \rightarrow \text{Spec } A$  is surjective.

*Proof.* For  $\mathfrak{p} \in \text{Spec } A$ , let  $S := A - \mathfrak{p}$  and denote  $S^{-1}B$  by  $B_{\mathfrak{p}}$ . Then we have  $A_{\mathfrak{p}} \hookrightarrow B_{\mathfrak{p}}$  and  $B_{\mathfrak{p}}$  is finite over  $A_{\mathfrak{p}}$ . Let  $\mathfrak{P}B_{\mathfrak{p}}$  be a maximal ideal of  $B_{\mathfrak{p}}$ . We claim that  $\mathfrak{P}B_{\mathfrak{p}} \cap A_{\mathfrak{p}}$  is maximal. Indeed, consider  $A_{\mathfrak{p}}/(\mathfrak{P} \cap A_{\mathfrak{p}}) \hookrightarrow B_{\mathfrak{p}}/\mathfrak{P}B_{\mathfrak{p}}$ , the latter is finite over the former. This enforces  $A_{\mathfrak{p}}/(\mathfrak{P}B_{\mathfrak{p}} \cap A_{\mathfrak{p}})$  to be a field. Hence  $\mathfrak{P}B_{\mathfrak{p}} \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$ , and then  $\mathfrak{P} \cap A = \mathfrak{p}$ .  $\square$

**Proposition 1.5.3.** Let  $A \subset B$  be noetherian rings such that  $B$  is finite over  $A$ . Then  $\dim A = \dim B$ .

*Proof.* If we have a sequence  $\mathfrak{P}_1 \subsetneq \mathfrak{P}_2$  of prime ideals in  $B$ , then there exists  $f \in \mathfrak{P}_2 \setminus \mathfrak{P}_1$ . Since  $B$  is finite over  $A$ , there exist  $a_1, \dots, a_n \in A$  such that

$$f^n + a_1 f^{n-1} + \dots + a_n = 0.$$

Then  $a_n \in \mathfrak{P}_2 \cap A$ . If  $a_n \in \mathfrak{P}_1$ ,  $f^{n-1} + \dots + a_{n-1}f \in \mathfrak{P}_1$  since  $f \notin \mathfrak{P}_1$ . Then  $a_{n-1} \in \mathfrak{P}_2$ . Repeat the process, it will terminate, whence  $\mathfrak{P}_1 \cap A \subsetneq \mathfrak{P}_2 \cap A$ . Otherwise, we have  $f^n \in a_1 B + \dots + a_n B \subset \mathfrak{P}_1$ .

Conversely, suppose we have  $\mathfrak{p}_1, \mathfrak{p}_2 \in \text{Spec } A$  with  $\mathfrak{p}_1 \subsetneq \mathfrak{p}_2$ . Choose  $\mathfrak{P}_1 \in \text{Spec } B$  such that  $\mathfrak{P}_1 \cap A = \mathfrak{p}_1$ , then we have  $A/\mathfrak{p}_1 \subset B/\mathfrak{P}_1$ . Let  $\mathfrak{P}_2$  be the preimage of the prime ideal in  $B/\mathfrak{P}_1$  which is over image of  $\mathfrak{p}_2$  in  $A/\mathfrak{p}_1$ . Proposition 1.5.2 guarantees that such  $\mathfrak{P}_2$  exists. Then we get  $\mathfrak{P}_1 \subsetneq \mathfrak{P}_2$ . Repeat this progress, we get  $\dim B \geq \dim A$ .  $\square$

Yang: To be completed

**Definition 1.5.4.** An integral domain  $A$  is called *normal* if it is integrally closed in its field of fractions  $\text{Frac}(A)$ .

**Lemma 1.5.5.** Let  $A \subset C$  be rings and  $B$  the integral closure of  $A$  in  $C$ ,  $S$  a multiplicatively closed subset of  $A$ . Then the integral closure of  $S^{-1}A$  in  $S^{-1}C$  is  $S^{-1}B$ .

*Proof.* For every  $b \in B$  and  $\forall s \in S$ , there exists  $a_i \in A$  s.t.

$$b^n + a_1 b^{n-1} + \dots + a_n = 0.$$

Then

$$\left(\frac{b}{s}\right)^n + \frac{a_1}{s^1} \left(\frac{b}{s}\right)^{n-1} + \cdots + \frac{a_n}{s^n} = 0.$$

Hence  $b/s$  is integral over  $S^{-1}A$ ,  $S^{-1}B$  is integral over  $S^{-1}A$ .

If  $c/s \in S^{-1}C$  is integral over  $S^{-1}A$ , then  $\exists a_i \in S^{-1}A$  s.t.

$$\left(\frac{c}{s}\right)^n + a_1 \left(\frac{c}{s}\right)^{n-1} + \cdots + a_n = 0.$$

Then

$$c^n + a_1 s c^{n-1} + \cdots + a_n s^n = 0 \in S^{-1}C$$

Then  $\exists t \in S$  s.t.

$$t(c^n + a_1 s c^{n-1} + \cdots + a_n s^n) = 0 \in C.$$

Then

$$(ct)^n + a_1 st(ct)^{n-1} + \cdots + a_n s^n t^n = t^n(c^n + a_1 s c^{n-1} + \cdots + a_n s^n) = 0.$$

Hence  $ct$  is integral over  $A$ , then  $ct \in B$ . Then  $c/s = (ct)/(st) \in S^{-1}B$ . This completes the proof.  $\square$

**Proposition 1.5.6.** Normality is a local property. That is, for an integral domain  $A$ , TFAE:

- (i)  $A$  is normal.
- (ii) For any prime ideal  $\mathfrak{p} \in \text{Spec } A$ , the localization  $A_{\mathfrak{p}}$  is normal.
- (iii) For any maximal ideal  $\mathfrak{m} \in \text{mSpec } A$ , the localization  $A_{\mathfrak{m}}$  is normal.

*Proof.* When  $A$  is normal,  $A_{\mathfrak{p}}$  is normal by Lemma 1.5.5.

Assume that  $A_{\mathfrak{m}}$  is normal for every  $\mathfrak{m} \in \text{mSpec } A$ . If  $A$  is not normal, let  $\tilde{A}$  be the integral closure of  $A$  in  $\text{Frac } A$ ,  $\tilde{A}/A$  is a nonzero  $A$ -module. Suppose  $\mathfrak{p} \in \text{Supp } \tilde{A}/A$  and  $\mathfrak{p} \subset \mathfrak{m}$ . We have  $\tilde{A}_{\mathfrak{m}}/A_{\mathfrak{m}} = 0$  and  $\tilde{A}_{\mathfrak{p}}/A_{\mathfrak{p}} = (\tilde{A}_{\mathfrak{m}}/A_{\mathfrak{m}})_{\mathfrak{p}} \neq 0$ . This is a contradiction.  $\square$

**Proposition 1.5.7.** Let  $A$  be a normal ring. Then  $A[X]$  is also normal.

**Definition 1.5.8.** A scheme  $X$  is called *normal* if the local ring  $\mathcal{O}_{X,\xi}$  is normal for any point  $\xi \in X$ . A ring  $A$  is called *normal* if  $\text{Spec } A$  is normal.

**Remark 1.5.9.** For a general ring  $A$ , let  $S := A \setminus (\bigcup_{\mathfrak{p} \in \text{Ass } A} \mathfrak{p}) = \bigcap_{\mathfrak{p} \in \text{Ass } A} A \setminus \mathfrak{p}$ . Then  $S$  is a multiplicative set. The localization  $S^{-1}A$  is called *the total ring of fractions* of  $A$ .

Suppose  $A$  is reduced and  $\text{Ass } A = \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$ . Denote its total ring of fractions by  $Q$ . Note that elements in  $Q$  are either unit or zero divisor. Hence any maximal ideal  $\mathfrak{m}$  is contained in  $\bigcup \mathfrak{p}_i Q$ , whence contained in some  $\mathfrak{p}_i Q$ . Thus  $\mathfrak{p}_i Q$  are maximal ideals. And we have  $\bigcap \mathfrak{p}_i Q = 0$ . By the Chinese Remainder Theorem, we have  $Q = \prod Q/\mathfrak{p}_i Q = \prod A_{\mathfrak{p}_i}$ .

Let  $A$  be a reduced ring with total ring of fractions  $Q$ . Then  $A$  is normal iff  $A$  is integral closed in  $Q$ . If  $A$  is normal, then for every  $\mathfrak{p} \in \text{Spec } A$ ,  $A_{\mathfrak{p}}$  is integral. Then there is unique minimal prime ideal  $\mathfrak{p}_i \subset \mathfrak{p}$ . In particular, any two minimal prime ideal are relatively prime. By the Chinese Remainder Theorem,  $A = \prod A/\mathfrak{p}_i$ . Just need to check  $A/\mathfrak{p}_i$  is integral closed in  $A_{\mathfrak{p}_i}$ . This is clear by check pointwise.

Conversely, suppose  $A$  is integral closed in  $Q$ . Let  $e_i$  be the unit element of  $A_{p_i}$ . It belongs to  $A$  since  $e_i^2 - e_i = 0$ . Since  $1 = e_1 + \cdots + e_n$  and  $e_i e_j = \delta_{ij}$ , we have  $A = \prod A e_i$ . Since  $A e_i$  is integral closed in  $A_{p_i}$ , it is normal. Hence  $A$  is normal.

**Lemma 1.5.10.** Let  $A$  be a normal ring. Then  $A$  verifies  $(R_1)$  and  $(S_2)$ .

*Proof.* Since all properties are local, we can assume  $A$  is integral and local.

For  $(S_2)$ , by Example ??, we only need to show that  $\text{Ass}_A A/fA$  has no embedded point. Let  $\mathfrak{p} = (f : g) \in \text{Ass}_A A/fA$  and  $t := f/g \in \text{Frac } A$ . After Replacing  $A$  by  $A_{\mathfrak{p}}$ , we can assume that  $\mathfrak{p}$  is maximal. By definition,  $t^{-1}\mathfrak{p} \subset A$ . If  $t^{-1}\mathfrak{p} \subset \mathfrak{p}$ , suppose  $\mathfrak{p}$  is generated by  $(x_1, \dots, x_n)$  and  $t^{-1}(x_1, \dots, x_n)^T = \Phi(x_1, \dots, x_n)^T$  for  $\Phi \in M_n(A)$ . There is a monic polynomial  $\chi(T) \in A[T]$  vanishing  $\Phi$ . Then  $\chi(t^{-1}) = 0$  and  $t^{-1} \in A$ . This is impossible by definition of  $t$ . Then  $t^{-1}\mathfrak{p} = A$ , and  $\mathfrak{p} = (t)$  is principal. By Krull's Principal Ideal Theorem 1.4.4,  $\text{ht}(\mathfrak{p}) = 1$ .

Now we show that  $A$  verifies  $(R_1)$ . Suppose  $(A, \mathfrak{m})$  is local of dimension 1. Choosing  $a \in \mathfrak{m}$ ,  $A/a$  is of dimension 0. Then by 1.3.8,  $\mathfrak{m}^n \subset aA$  for some  $n \geq 1$ . Suppose  $\mathfrak{m}^{n-1} \not\subset aA$ . Choose  $b \in \mathfrak{m}^{n-1} \setminus aA$  and let  $t = a/b$ . By construction,  $t^{-1} \notin A$  and  $t^{-1}\mathfrak{m} \subset A$ . After similar argument, we see that  $\mathfrak{m} = tA$ , whence  $A$  is regular.  $\square$

**Lemma 1.5.11.** Let  $(A, \mathfrak{m})$  be a noetherian local ring of dimension 1. Then  $A$  is normal iff  $A$  is regular.

*Proof.* By lemma 1.5.10, we just need to show that regularity implies normality.

Let  $t \in \mathfrak{m} \setminus \mathfrak{m}^2$ . Since  $A$  is regular,  $\mathfrak{m} = (t)$ . Let  $I \subset \mathfrak{m}$  be an ideal. If  $I \subset \bigcap_n \mathfrak{m}^n$ , then for every  $a \in I$ , there exists  $a_n$  such that  $a = a_n t^n$ . Then we get an ascending chain of ideals  $(a_1) \subset (a_2) \subset \cdots$ . Hence  $a = 0$  by Nakayama's Lemma. Suppose  $I$  is not zero. Then there is some  $n$  such that  $I \subset \mathfrak{m}^n$  and  $I \not\subset \mathfrak{m}^{n+1}$ . For every  $at^n \in I \setminus \mathfrak{m}^{n+1}$ ,  $a \notin \mathfrak{m}$ , whence  $a$  is a unit in  $A$ . Then  $I = (t^n)$ . Hence  $A$  is PID and hence normal.  $\square$

**Proposition 1.5.12.** Let  $A$  be a noetherian integral domain of dimension  $\geq 1$  verifying  $(S_2)$ . Then

$$A = \bigcap_{\mathfrak{p} \in \text{Spec } A, \text{ht}(\mathfrak{p})=1} A_{\mathfrak{p}}.$$

*Proof.* Clearly  $A \subset \bigcap A_{\mathfrak{p}}$ . Let  $t = f/g \in \bigcap A_{\mathfrak{p}}$ . Since  $f \in gA_{\mathfrak{p}}$  and we have  $gA = \bigcap (gA_{\mathfrak{p}} \cap A)$ ,  $f \in gA$ . It follows that  $t \in A$ .  $\square$

**Theorem 1.5.13** (Serre's criterion for normality). Let  $X$  be a locally noetherian scheme. Then  $X$  is normal if and only if it verifies  $(R_1)$  and  $(S_2)$ .

*Proof.* One direction has been proved in Lemma 1.5.10. Suppose  $X$  verifies  $(R_1)$  and  $(S_2)$ . Again we can assume  $X = \text{Spec } A$  is affine and  $A$  is local. By Remark 1.5.9, we just need to show that  $A$  is integral closed in its total ring of fractions  $Q$ . Suppose we have

$$\left(\frac{a}{b}\right)^n + c_1 \left(\frac{a}{b}\right)^{n-1} + \cdots + c_n = 0 \in Q.$$

Since  $A$  verifies  $(S_2)$ ,  $bA = \bigcap \mathfrak{v}_{\mathfrak{p}}^{-1}(b_{\mathfrak{p}}A_{\mathfrak{p}})$ . So it is sufficient to show that  $a_{\mathfrak{p}} \in b_{\mathfrak{p}}A_{\mathfrak{p}}$  with  $\text{ht}(\mathfrak{p}) = 1$ . Note that  $A_{\mathfrak{p}}$  is regular and hence normal by Lemma 1.5.11. Then above equation gives us desired result.  $\square$

## 1.6 Smoothness

### 1.6.1 Modules of differentials and derivations

In this subsection, let  $R$  be a ring and  $A$  an  $R$ -algebra.

**Definition 1.6.1** (Derivation). A *derivation* of  $A$  over  $R$  is an  $R$ -linear map  $\partial : A \rightarrow M$  with an  $A$ -module such that for all  $a, b \in A$ , we have

$$\partial(ab) = a\partial(b) + b\partial(a).$$

Given the module  $M$ , the set of all derivations of  $A$  over  $R$  into  $M$  forms an  $A$ -module, denoted by  $\text{Der}_R(A, M)$ .

Given a module homomorphism  $f : M \rightarrow N$  of  $A$ -modules and a derivation  $\partial \in \text{Der}_R(A, M)$ , the map  $f \circ \partial$  is a derivation of  $A$  over  $R$  into  $N$ . This gives a functor  $\text{Der}_R(A, -) : \mathbf{Mod}_A \rightarrow \mathbf{Mod}_A$ .

**Proposition 1.6.2.** The functor  $\text{Der}_R(A, -)$  is representable. The representing object is denoted by  $(\Omega_{A/R}, \Psi)$ , where  $\Psi : \text{Hom}_A(\Omega_{A/R}, -) \rightarrow \text{Der}_R(A, -)$  is given by pullback along a universal  $R$ -derivation  $d_A : A \rightarrow \Omega_{A/R}$ . The module  $\Omega_{A/R}$  is called the *module of differentials* of  $A$  over  $R$ .

*Proof.* First suppose  $A$  is a free  $R$ -algebra with a set of generators  $a_\lambda, \lambda \in \Lambda$ . Then an  $R$ -derivation  $\partial \in \text{Der}_R(A, M)$  is uniquely determined by its values on the generators  $a_\lambda$ . Let

$$\Omega_{A/R} := \bigoplus_{\lambda \in \Lambda} A \cdot da_\lambda$$

and  $d : A \rightarrow \Omega_{A/R}$  be the  $R$ -derivation defined by  $a_\lambda \mapsto da_\lambda$ . For any  $R$ -derivation  $\partial \in \text{Der}_R(A, M)$ , we can define a unique  $A$ -module homomorphism  $\Phi_\partial : \Omega_{A/R} \rightarrow M$  by sending  $da_\lambda$  to  $\partial(a_\lambda)$  such that  $\partial = \Phi_\partial \circ d$ . This gives a bijection

$$\text{Der}_R(A, M) \cong \text{Hom}_A(\Omega_{A/R}, M), \quad \partial \mapsto \Phi_\partial.$$

Now suppose  $A = F/I$  is an arbitrary  $R$ -algebra, where  $F$  is a free  $R$ -algebra and  $I$  is an ideal of  $F$ . Then we can define the module of differentials

$$\Omega_{A/R} := (\Omega_{F/R} \otimes_F A) / \sum_{f \in I} A \cdot df.$$

The  $R$ -linear map  $d_A : F \otimes_F A \xrightarrow{d_F} \Omega_{F/R} \otimes_F A \rightarrow \Omega_{A/R}$  is a derivation of  $A$  over  $R$ .

For any  $R$ -derivation  $\partial \in \text{Der}_R(A, M)$ , note that  $F \rightarrow A \xrightarrow{\partial} M$  is an  $R$ -derivation of  $F$  over  $R$  into  $M$ . Then we get an  $F$ -module homomorphism  $\Omega_F \rightarrow M$ . It gives an  $A$ -module homomorphism  $\Omega_F \otimes_F A \rightarrow M, df \otimes 1 \mapsto \partial f$ . This map factors into  $\Omega_F \otimes_F A \rightarrow \Omega_{A/R}$  and  $\Phi_\partial : \Omega_{A/R} \rightarrow M$ . Since  $\Phi_\partial$  is  $A$ -linear and  $\Omega_{A/R}$  is generated by  $da_\lambda$  as  $A$ -module, such  $\Phi_\partial$  is unique.  $\square$

**Corollary 1.6.3.** Suppose  $A$  is of finite type over  $R$ . Then the module of differentials  $\Omega_{A/R}$  is a finitely generated  $A$ -module.

**Remark 1.6.4.** Let  $B$  be an  $A$ -algebra,  $M$  an  $A$ -module and  $N$  a  $B$ -module. If there is a homomorphism of  $A$ -modules  $M \rightarrow N$ , then we can extend it to a homomorphism of  $B$ -modules  $M \otimes_A B \rightarrow N$  by sending  $m \otimes b$  to  $m \cdot b$ . And such extension is unique in the sense of following commutative diagram:

$$\begin{array}{ccc} M & \xrightarrow{\quad} & N \\ \downarrow & \nearrow \exists! & \\ M \otimes_A B & & \end{array}$$

Hence we get a natural bijection

$$\mathrm{Hom}_A(M, N) \cong \mathrm{Hom}_B(M \otimes_A B, N).$$

**Proposition 1.6.5.** Let  $A, R'$  be  $R$ -algebras and  $A' := A \otimes_R R'$ . Then the module of differentials  $\Omega_{A'/R'}$  is isomorphic to  $\Omega_{A/R} \otimes_A A'$ .

*Proof.* We check the universal property of  $\Omega_{A/R} \otimes_A A'$ . First, the map

$$d_{A'} : A \otimes_R R' \rightarrow \Omega_{A/R} \otimes_R R' \cong \Omega_{A/R} \otimes_A A', \quad a \otimes r \mapsto da \otimes r$$

is an  $R'$ -derivation of  $A'$  into  $\Omega_{A/R} \otimes_A A'$ . For any  $R'$ -derivation  $\partial' : A' \rightarrow M$  into an  $A'$ -module  $M$ , we can compose it with the homomorphism  $A' \rightarrow A$  and get an  $R$ -derivation  $\partial : A \rightarrow M$ . By the universal property of  $\Omega_{A/R}$ , there is a unique  $A$ -module homomorphism  $\Phi : \Omega_{A/R} \rightarrow M$  such that  $\partial = \Phi \circ d_A$ . Then we can extend it to an  $A'$ -module homomorphism  $\Phi' : \Omega_{A/R} \otimes_A A' \rightarrow M$  by Remark 1.6.4. By the construction, we have  $\Phi' \circ d_{A'} = \partial'$ .  $\square$

**Proposition 1.6.6.** Let  $A$  be an  $R$ -algebra and  $S$  a multiplicative set of  $A$ . Then we have an isomorphism

$$\Omega_{S^{-1}A/R} \cong S^{-1}\Omega_{A/R}.$$

*Proof.* Let

$$d_{S^{-1}A} : S^{-1}A \rightarrow S^{-1}\Omega_{A/R}, \quad \frac{a}{s} \mapsto \frac{sda - ads}{s^2}.$$

By direct computation,  $d_{S^{-1}A}$  is an  $R$ -derivation of  $S^{-1}A$  over  $R$  into  $S^{-1}\Omega_{A/R}$ . For any  $R$ -derivation  $\partial : S^{-1}A \rightarrow M$  into an  $S^{-1}A$ -module  $M$ , we can get an  $S^{-1}A$ -module homomorphism  $\Phi' : S^{-1}\Omega_{A/R} \rightarrow M$  as proof of Proposition 1.6.5. We have

$$\partial(s \cdot \frac{a}{s}) = s\partial(\frac{a}{s}) + \frac{a}{s}\partial s.$$

It follows that

$$\partial(\frac{a}{s}) = \frac{s\partial a - a\partial s}{s^2} = \frac{s\Phi'(da) - a\Phi'(ds)}{s^2} = \Phi'(\frac{sda - ads}{s^2}).$$

Thus,  $\Phi' \circ d_{S^{-1}A} = \partial$ .  $\square$

**Theorem 1.6.7.** Let  $A$  be an  $R$ -algebra and  $B$  an  $A$ -algebra. Then there is a natural short exact sequence

$$\Omega_{A/R} \otimes_A B \rightarrow \Omega_{B/R} \rightarrow \Omega_{B/A} \rightarrow 0$$

of  $B$ -modules.

*Proof.* Let  $d_{A/R} : A \rightarrow \Omega_{A/R}$  be the  $R$ -derivation of  $A$  over  $R$ . The map  $A \rightarrow B \xrightarrow{d_{B/R}} \Omega_{B/R}$  induces a  $B$ -linear map

$$u : \Omega_{A/R} \otimes_A B \rightarrow \Omega_{B/R}, \quad d_{A/R}(a) \otimes b \mapsto bd_{B/R}(a).$$

The map  $d_{B/A}$  is an  $A$ -derivation and hence  $R$ -derivation. Then it induces a  $B$ -linear map

$$v : \Omega_{B/R} \rightarrow \Omega_{B/A}, \quad d_{B/R}(b) \mapsto d_{B/A}(b).$$

Since  $\Omega_{B/A}$  is generated by elements of the form  $d_{B/A}(b)$  for  $b \in B$ , the map  $v$  is surjective. And clearly  $d_{B/A}(a) = ad_{B/A}(1) = 0$  for  $a \in A$ .

Consider the composition  $B \xrightarrow{d_{B/R}} \Omega_{B/R} \rightarrow \Omega_{B/R}/\mathfrak{S}u$ . For every  $a \in A, b \in B$ , we have

$$[d_{B/R}(ab)] = [bd_{B/R}(a) + ad_{B/R}(b)] = [bd_{B/R}(a)] + [ad_{B/A}(b)] = [ad_{B/A}(b)].$$

Hence it is indeed an  $A$ -derivation of  $B$ . Then it induces a  $B$ -linear map

$$\varphi : \Omega_{B/A} \rightarrow \Omega_{B/R}/\mathfrak{S}u, \quad d_{B/A}(b) \mapsto [d_{B/R}(b)].$$

The map  $\varphi$  is surjective since  $\Omega_{B/R}$  is generated by elements of the form  $d_{B/R}(b)$  for  $b \in B$ . Note that the composition

$$\Omega_{B/A} \xrightarrow{\varphi} \Omega_{B/R}/\mathfrak{S}u \rightarrow \Omega_{B/A}/\text{Ker } v$$

is the identity map. Thus,  $\varphi$  is injective and hence an isomorphism. In particular, we have  $\text{Ker } v = \mathfrak{S}u$ .  $\square$

**Remark 1.6.8.** The exact sequence in Theorem 1.6.7 is left exact if and only if every  $R$ -derivation of  $A$  into  $B$ -module extends to an  $R$ -derivation of  $B$  into  $B$ -module.

Yang: To be completed.

**Theorem 1.6.9.** Let  $A$  be an  $R$ -algebra and  $I$  an ideal of  $A$ . Set  $B := A/I$ . Then there is a natural short exact sequence

$$I/I^2 \rightarrow \Omega_{A/R} \otimes_A B \rightarrow \Omega_{B/R} \rightarrow 0$$

of  $B$ -modules.

*Proof.* Suppose  $A = F/\mathfrak{b}$  for some free  $R$ -algebra  $F$  and an ideal  $\mathfrak{b}$  of  $F$ . Let  $\mathfrak{a}$  be the preimage of  $I$  in  $F$ . Let  $d\mathfrak{b}$  (resp.  $d\mathfrak{a}$ ) denote the image of  $\mathfrak{b}$  (resp.  $\mathfrak{a}$ ) in  $\Omega_{F/R}$ . Then we have

$$\Omega_{A/R} \otimes_A B = \Omega_{F/R} \otimes_F B / (d\mathfrak{b} \otimes_F B), \quad \Omega_{B/R} = \Omega_{F/R} \otimes_F B / (d\mathfrak{a} \otimes_F B).$$

Clearly

$$I/I^2 \cong (\mathfrak{a}/\mathfrak{b}) \otimes_F B \rightarrow (d\mathfrak{a} \otimes_F B) / (d\mathfrak{b} \otimes_F B)$$

is surjective. Then the exact sequence follows.  $\square$

**Definition 1.6.10.** Let  $\mathbf{k}$  be a field and  $A$  an integral  $\mathbf{k}$ -algebra of finite type of dimension  $n$ . We say  $A$  is *smooth at*  $\mathfrak{p} \in \text{Spec } A$  if the module of differentials  $\Omega_{A,\mathfrak{p}}$  is a free  $A_{\mathfrak{p}}$ -module of rank  $n$ .

**Example 1.6.11.** Let  $\mathbf{K}/\mathbf{k}$  be a finite generated field extension and  $\mathbf{k}'$  be the algebraic closure of  $\mathbf{k}$



in  $\mathbf{K}$ . Then

$$\dim_{\mathbf{K}} \Omega_{\mathbf{K}/\mathbf{k}} = \text{trdeg}(\mathbf{K}/\mathbf{k}) + \dim_{\mathbf{k}'} \Omega_{\mathbf{k}'/\mathbf{k}},$$

and  $\dim_{\mathbf{k}'} \Omega_{\mathbf{k}'/\mathbf{k}} = 0$  if and only if  $\mathbf{k}'$  is separable over  $\mathbf{k}$ .

**Yang:** The conclusion is totally wrong.

First suppose  $\mathbf{K} = \mathbf{k}'$  is algebraic over  $\mathbf{k}$ . Suppose  $\mathbf{k}'/\mathbf{k}$  is separable. For every  $\alpha \in \mathbf{k}'$ , suppose  $f(\alpha) = 0$  for  $f \in \mathbf{k}[T]$ . Then  $df(\alpha) = f'(\alpha)d\alpha = 0$ . By the separability of  $\mathbf{k}'/\mathbf{k}$ , we have  $f'(\alpha) \neq 0$ . It follows that  $d\alpha = 0$ . Conversely, let  $\alpha \in \mathbf{k}'$  be an inseparable element over  $\mathbf{k}$ . Since  $\mathbf{k}[\alpha] \rightarrow \mathbf{k}[\alpha], \alpha^n \mapsto n\alpha^{n-1}$  is a non-zero  $R$ -derivation, we have  $\Omega_{\mathbf{k}[\alpha]/\mathbf{k}} \neq 0$ . By induction on number of generated elements, choosing a middle field  $\mathbf{k} \subset \mathbf{k}'' \subset \mathbf{k}'$ , at least one of  $\Omega_{\mathbf{k}''/\mathbf{k}}$  and  $\Omega_{\mathbf{k}'/\mathbf{k}''}$  is non-zero. Then  $\Omega_{\mathbf{K}/\mathbf{k}} \neq 0$  by Theorem 1.6.7.

Then suppose  $\mathbf{k}' = \mathbf{k}$ . By the Noether's Normalization Lemma, we can find a finite set of elements  $T_1, \dots, T_n \in \mathbf{K}$  such that  $\mathbf{K}$  is algebraic over  $\mathbf{k}'(T_1, \dots, T_n)$ . Note that we can choose  $T_i$  such that  $\mathbf{K}/\mathbf{k}'(T_1, \dots, T_n)$  is separable. To see this, if  $\alpha \in \mathbf{K}$  is an inseparable element over  $\mathbf{k}'(T_1, \dots, T_n)$ , then by replacing a suitable  $T_i$  with  $\alpha$ , we reduce the inseparable degree of  $\mathbf{K}/\mathbf{k}'(T_1, \dots, T_n)$ .

Since  $\mathbf{K}/\mathbf{k}'(T_1, \dots, T_n)$  is finite, every  $\mathbf{k}$ -derivation of  $\mathbf{k}'(T_1, \dots, T_n)$  into  $\mathbf{K}$ -module extends to a  $\mathbf{k}$ -derivation of  $\mathbf{K}$  into  $\mathbf{K}$ -module. Then by Remark 1.6.8, we have

$$0 \rightarrow \Omega_{\mathbf{k}'(T_1, \dots, T_n)/\mathbf{k}} \otimes_{\mathbf{k}'(T_1, \dots, T_n)} \mathbf{K} \rightarrow \Omega_{\mathbf{K}/\mathbf{k}} \rightarrow \Omega_{\mathbf{K}/\mathbf{k}'(T_1, \dots, T_n)} \rightarrow 0.$$

Finally, note that every  $\mathbf{k}$ -derivation  $\partial$  of  $\mathbf{k}'$  into  $\mathbf{K}$ -module can be extended to  $\mathbf{k}'[T_1, \dots, T_n]$  by setting  $\partial T_i = 0$ . Thus, we have

$$0 \rightarrow \Omega_{\mathbf{k}'/\mathbf{k}} \otimes_{\mathbf{k}'} \mathbf{k}'[T_1, \dots, T_n] \rightarrow \Omega_{\mathbf{k}'[T_1, \dots, T_n]/\mathbf{k}} \rightarrow \Omega_{\mathbf{k}'[T_1, \dots, T_n]/\mathbf{k}'} \rightarrow 0.$$

This follows that

$$\dim_{\mathbf{K}} \Omega_{\mathbf{K}/\mathbf{k}} = \dim_{\mathbf{K}} \Omega_{\mathbf{K}/\mathbf{k}'} + \dim_{\mathbf{k}'} \Omega_{\mathbf{k}'/\mathbf{k}}.$$

## 1.6.2 Applications to affine varieties

Let  $\mathbf{k}$  be arbitrary field,  $A = \mathbf{k}[T_1, \dots, T_n]$  and  $\mathfrak{m}$  a maximal ideal of  $A$  such that  $\kappa(\mathfrak{m})$  is separable over  $\mathbf{k}$ . We try to give an explanation of Zariski's tangent space at  $\mathfrak{m}$  using the language of derivation. We know that  $\Omega_{A/\mathbf{k}} = \bigoplus_{i=1}^n A dT_i$ , thus  $\Omega_{A_{\mathfrak{m}}/\mathbf{k}} \cong \bigoplus_{i=1}^n A_{\mathfrak{m}} dT_i$ . Then

$$\text{Der}_{\mathbf{k}}(A_{\mathfrak{m}}, A_{\mathfrak{m}}) \cong \text{Hom}_{\mathbf{k}}(\Omega_{A_{\mathfrak{m}}/\mathbf{k}}, A_{\mathfrak{m}}) \cong \bigoplus_{i=1}^n A_{\mathfrak{m}} \partial_i,$$

where  $\partial_i \in \text{Der}_{\mathbf{k}}(A_{\mathfrak{m}}, A_{\mathfrak{m}})$  is the derivation defined by  $dT_i \mapsto 1$  and  $dT_j \mapsto 0$  for  $j \neq i$ . It coincides with the usual derivation  $f \mapsto \partial f / \partial T_i$ . Consider the restriction of  $\partial_i$  to  $\mathfrak{m}$  and take values in the residue field  $\kappa(\mathfrak{m})$ , we get

$$\Phi : \mathfrak{m} \xrightarrow{(\partial_1, \dots, \partial_n)^T} A_{\mathfrak{m}}^n \rightarrow \kappa(\mathfrak{m})^n.$$

Since  $\kappa(\mathfrak{m})$  is separable over  $\mathbf{k}$ , we claim that  $\text{Ker } \Phi = \mathfrak{m}^2$ . Indeed, by Remark 1.6.12, we can write every  $f \in \mathfrak{m} \setminus \mathfrak{m}^2$  as  $\sum_i a_i g_i$ . Then

$$\frac{\partial f}{\partial T_i} = a_i \frac{\partial g_i}{\partial T_i} + g_i \frac{\partial a_i}{\partial T_i}.$$



Since  $g_i$  is separable, the image of  $\partial g_i / \partial T_i$  in  $\kappa(\mathfrak{m})$  is not zero. Hence  $\Phi(f) \neq 0$ . By the claim,  $\Phi$  induces an isomorphism  $\mathfrak{m}/\mathfrak{m}^2 \cong \kappa(\mathfrak{m})^n$  of  $\kappa(\mathfrak{m})$ -vector spaces. Then we get

$$T_{A,\mathfrak{m}} = (\mathfrak{m}/\mathfrak{m}^2)^\vee \cong \bigoplus_{i=1}^n \kappa(\mathfrak{m}) \cdot \partial_i|_x,$$

where  $x \in \mathfrak{a}_{\mathbf{k}}^n$  is the point corresponding to  $\mathfrak{m}$ . This coincides with the usual tangent space at  $x$  in language of differential geometry.

**Remark 1.6.12.** Let  $\mathbf{k}$  be arbitrary field,  $A = \mathbf{k}[T_1, \dots, T_n]$  and  $g_i$  irreducible polynomials in one variable  $T_i$  over  $\mathbf{k}$ . Then for every  $f \in A$ , we can write

$$f = \sum_{I=(i_1, \dots, i_n) \in \mathbb{Z}_{\geq 0}^n} a_I g_1^{i_1} \cdots g_n^{i_n}, \quad a_I \in A, \quad \deg_{T_i} a_I \leq \deg g_i.$$

This is called the *Taylor expansion of  $f$  with respect to  $g_1, \dots, g_n$* .

When  $n = 1$ , it follows from division algorithm. For  $n > 1$ , we can use induction on  $n$ . Let  $\mathbf{K} = \mathbf{k}(T_1, \dots, T_{n-1})$ . Then we can write  $f$  as

$$f = \sum_{i=0}^r a_i g_n^i, \quad a_i \in \mathbf{K}[T_n], \quad \deg a_i < \deg g_n.$$

Comparing the coefficients of two sides from the highest degree of  $T_n$  to the lowest degree, we see that

$$a_i \in \mathbf{k}[T_1, \dots, T_{n-1}].$$

By induction hypothesis, the conclusion follows.

Let  $B = A/I$  be a  $\mathbf{k}$  of finite type,  $I = (F_1, \dots, F_m) \subset \mathfrak{m}$  and  $\mathfrak{n}$  the image of  $\mathfrak{m}$  in  $B$ . We have an exact sequence of  $\kappa(\mathfrak{m})$ -vector spaces

$$0 \rightarrow I/(I \cap \mathfrak{m}^2) \rightarrow \mathfrak{m}/\mathfrak{m}^2 \rightarrow \mathfrak{n}/\mathfrak{n}^2 \rightarrow 0.$$

It induces an isomorphism

$$T_{B,\mathfrak{n}} \cong \{\partial \in T_{A,\mathfrak{m}} : \partial(f) = 0, \forall f \in I\}.$$

The *Jacobian matrix* of  $F_1, \dots, F_m$  is the  $m \times n$  matrix

$$J(F_1, \dots, F_m) := \left( \frac{\partial F_i}{\partial T_j} \right)_{1 \leq i \leq m, 1 \leq j \leq n}$$

with entries in  $B$ .

**Theorem 1.6.13.** Setting as above. Then  $B$  is regular at  $\mathfrak{n}$  if and only if the Jacobian matrix  $J$  has maximal rank  $n - \dim B_{\mathfrak{n}}$  after taking values in the residue field  $\kappa(\mathfrak{m})$ .

*Proof.* We have an exact sequence

$$0 \rightarrow T_{B,\mathfrak{n}} \rightarrow T_{A,\mathfrak{m}} \xrightarrow{\Psi} \kappa^m \rightarrow 0,$$

where  $\Psi$  sends  $\partial \in T_{A,\mathfrak{m}}$  to  $(\partial(F_1), \dots, \partial(F_m))^T$ . Note that the matrix of  $\Psi$  is just  $J^T$ , the transpose of the Jacobian matrix. Hence

$$\text{rank } J = n - \dim_{\kappa} T_{B,\mathfrak{n}} \leq n - \dim B_{\mathfrak{n}}$$

and the equality holds if and only if  $B$  is regular at  $\mathfrak{n}$ .  $\square$

**Remark 1.6.14.** If  $\kappa(\mathfrak{m})$  is not separable over  $\mathbf{k}$ , then we still have the inequality

$$\text{rank } J \leq n - \dim B_{\mathfrak{n}}.$$

Indeed, in any case, we have an exact sequence

$$0 \rightarrow I/(I \cap \mathfrak{m}^2) \rightarrow \mathfrak{m}/\mathfrak{m}^2 \rightarrow \mathfrak{n}/\mathfrak{n}^2 \rightarrow 0.$$

Hence  $\dim_{\kappa} I/(I \cap \mathfrak{m}^2) = n - \dim B_{\mathfrak{n}}$ . There is a  $\kappa(\mathfrak{m})$ -linear map

$$I/(I \cap \mathfrak{m}^2) \rightarrow \kappa(\mathfrak{m})^n, \quad [f] \mapsto (\partial_1(f), \dots, \partial_n(f))^T,$$

and every row of the Jacobian matrix  $J$  is in the image of this map. Thus, the rank of  $J$  is at most  $n - \dim B_{\mathfrak{n}}$ .

Hence if  $\text{rank } J = n - \dim B_{\mathfrak{n}}$ , we can still see that  $B$  is regular at  $\mathfrak{n}$ . However, the converse does not hold in general.

**Proposition 1.6.15.** Let  $\mathbf{k}$  be a field,  $\mathbb{k}$  the algebraic closure of  $\mathbf{k}$ ,  $A$  a  $\mathbf{k}$ -algebra of finite type and  $A_{\mathbb{k}} := A \otimes_{\mathbf{k}} \mathbb{k}$ . **Yang: Suppose  $A_{\mathbb{k}}$  is integral.** Let  $\mathfrak{m} \in \text{mSpec } A$  and  $\mathfrak{m}'$  be a maximal ideal of  $A_{\mathbb{k}}$  lying over  $\mathfrak{m}$ . Then

- (a) If  $A_{\mathbb{k}}$  is regular at  $\mathfrak{m}'$ , then  $A$  is regular at  $\mathfrak{m}$ ;
- (b) suppose  $\kappa(\mathfrak{m})$  is separable over  $\mathbf{k}$ , the converse holds.

*Proof.* Regarding  $J_{\mathfrak{m}}$  and  $J_{\mathfrak{m}'}$  as matrices with entries in  $\mathbb{k}$ , they are the same and hence have the same rank. If  $A_{\mathbb{k}}$  is regular at  $\mathfrak{m}'$ , since  $\kappa(\mathfrak{m}) = \mathbb{k}$ , then  $\text{rank } J_{\mathfrak{m}'} = n - \dim A_{\mathbb{k}, \mathfrak{m}'}$ . Note that  $\dim A_{\mathbb{k}, \mathfrak{m}'} = \text{trdeg}(\mathcal{K}(A_{\mathbb{k}})/\mathbb{k}) = \text{trdeg}(\mathcal{K}(A)/\mathbf{k}) = \dim A_{\mathfrak{m}}$ , we have  $\text{rank } J_{\mathfrak{m}} = n - \dim A_{\mathfrak{m}}$ . Hence  $A$  is regular at  $\mathfrak{m}$ .

Conversely, suppose  $A$  is regular at  $\mathfrak{m}$  and  $\kappa(\mathfrak{m})$  is separable over  $\mathbf{k}$ . Then  $\text{rank } J_{\mathfrak{m}} = n - \dim A_{\mathfrak{m}}$ . Hence  $A_{\mathbb{k}}$  is regular at  $\mathfrak{m}'$ . **Yang: To be modified.**  $\square$

**Proposition 1.6.16.** Let  $\mathbf{k}$  be a field and  $A$  an integral  $\mathbf{k}$ -algebra of finite type and of dimension  $n$ . Let  $\mathbb{k}$  be the algebraic closure of  $\mathbf{k}$  and  $A_{\mathbb{k}} := A \otimes_{\mathbf{k}} \mathbb{k}$ . Then  $A$  is smooth at  $\mathfrak{p} \in \text{Spec } A$  if and only if  $A_{\mathbb{k}}$  is regular at every  $\mathfrak{m}'$  over  $\mathfrak{m}$ .

*Proof.* Since  $\Omega_{A_{\mathbb{k}}/\mathbb{k}} \cong \Omega_{A/\mathbf{k}} \otimes_A A_{\mathbb{k}}$  is free of rank  $n$  if and only if  $\Omega_{A/\mathbf{k}}$  is free of rank  $n$ , we can assume that  $\mathbf{k} = \mathbb{k}$ . If  $A$  is smooth at  $\mathfrak{p}$ , then  $\Omega_{A_{\mathfrak{p}}/\mathbf{k}} \cong \bigoplus A_{\mathfrak{p}} d\mathbf{f}_i$  is free of rank  $n$ . Let  $\mathfrak{P}_i \in \text{Der}_{\mathbf{k}}(A_{\mathfrak{m}}, A_{\mathfrak{m}})$  be the derivation defined by  $d\mathbf{f}_i \mapsto 1$  and  $d\mathbf{T}_j \mapsto 0$  for  $j \neq i$ . Then we have  $\partial_i \mathbf{f}_j = \delta_{ij}$  for  $1 \leq i, j \leq n$ . Then similar to above argument, we have an isomorphism

$$\mathfrak{m}/\mathfrak{m}^2 \xrightarrow{(\partial_1, \dots, \partial_n)^T} \mathbb{k}^n.$$

This shows that  $A_{\mathbb{k}}$  is regular at  $\mathfrak{m}$ .

Conversely, suppose  $A_{\mathbb{k}}$  is regular at  $\mathfrak{m}$ . Note that  $\mathfrak{m}/\mathfrak{m}^2 \rightarrow \Omega_{A, \mathbb{k}} \otimes_A \mathbb{k}$  is surjective since  $\Omega_{A_{\mathbb{k}}/\mathbb{k}} = 0$ . Then by Nakayama's lemma,  $\Omega_{A_{\mathfrak{m}}/\mathbf{k}}$  is generated by  $n$  elements as an  $A_{\mathfrak{m}}$ -module.

Note that  $\dim_{\mathcal{K}(A)} \Omega_{\mathcal{K}(A)/\mathbf{k}} = \text{trdeg}(\mathcal{K}(A)/\mathbf{k}) = \dim A_{\mathfrak{m}} = n$ . **Yang: By induction on transcendental degree.**

Yang: By Nakayama's Lemma,  $\Omega_{A_{\mathfrak{m}}/\mathbf{k}}$  is free of rank  $n$  as an  $A_{\mathfrak{m}}$ -module.

Yang: To be completed. □

**Example 1.6.17.** Let  $\mathbf{k}$  be an imperfect field of characteristic  $p > 2$ . Suppose  $\alpha = \beta^p \in \mathbf{k}$  and  $\beta$  is not in  $\mathbf{k}$ . Let  $A = \mathbf{k}[x, y]/(x^2 - y^p - \alpha)$  and  $\mathfrak{m} = (x, y^p - \alpha) = (x)$ . Note that  $\mathfrak{m}$  is principal, so  $A$  is regular at  $\mathfrak{m}$ . However,

$$J_{\mathfrak{m}} = \left( \frac{\partial}{\partial x}(x^2 - y^p - \alpha), \frac{\partial}{\partial y}(y^p - \alpha) \right) = (2x, 0) = (0, 0) \in M_{1 \times 2}(\kappa(\mathfrak{m})).$$

Thus,  $A$  is not smooth at  $\mathfrak{m}$ . From the view of differentials, we have

$$\Omega_{A_{\mathfrak{m}}/\mathbf{k}} = A_{\mathfrak{m}}dx \oplus A_{\mathfrak{m}}dy/A_{\mathfrak{m}} \cdot xdx = \kappa(\mathfrak{m})dx \oplus A_{\mathfrak{m}}dy,$$

which is not free as an  $A_{\mathfrak{m}}$ -module.

## 1.7 Formal Completion

### 1.7.1 Formal completion of rings and modules

**Definition 1.7.1.** Let  $A$  be a ring and  $\mathcal{T}$  a topology on  $A$ . We say that  $(A, \mathcal{T})$  is a *topological ring* if the operations of addition and multiplication are continuous with respect to the topology  $\mathcal{T}$ .

Given a topological ring  $A$ . A *topological  $A$ -module* is a pair  $(M, \mathcal{T}_M)$  where  $M$  is an  $A$ -module and  $\mathcal{T}_M$  is a topology on  $M$  such that the addition and scalar multiplication is continuous. The morphisms of topological  $A$ -modules are the continuous  $A$ -linear maps. They form a category denoted by **TopMod $_A$** .

**Definition 1.7.2.** Let  $A$  be a ring,  $I$  an ideal of  $A$  and  $M$  an  $A$ -module. The  *$I$ -adic topology* on  $M$  is the topology defined by the basis of open sets  $x + I^k M$  for all  $x \in M, k \geq 0$ .

**Example 1.7.3.** Let  $A = \mathbb{Z}$  be the ring of integers and  $p$  a prime number. The  $p$ -adic topology on  $\mathbb{Z}$  is defined by the metric

$$d(x, y) := \|x - y\|_p := p^{-v(x-y)},$$

where  $v$  is the valuation defined by the ideal  $p\mathbb{Z}$ .

Note that for  $I$ -adic topology, any homomorphism  $f : M \rightarrow N$  of  $A$ -modules is continuous since  $f(x + I^k N) \subset f(x) + I^k M$  for all  $x \in M$  and  $k \geq 0$ . Yang: Hence the forgotten functor **TopMod $_A$**   $\rightarrow$  **Mod $_A$**  gives an equivalence of categories.

Let  $M$  be an  $A$ -module equipped with the  $I$ -adic topology. Note that  $M$  is Hausdorff as a topological space if and only if  $\bigcap_{n \geq 0} I^n M = \{0\}$ . In this case, we say that  $M$  is  *$I$ -adically separated*.

When  $M$  is  $I$ -adically separated, we can see that  $M$  is indeed a metric space. Fix  $r \in (0, 1)$ . For every  $x \neq y \in M$ , there is a unique  $k \geq 0$  such that  $x - y \in I^k M$  but  $x - y \notin I^{k+1} M$ . We can define a metric on  $M$  by

$$d(x, y) := r^k.$$

This metric induces the  $I$ -adic topology on  $M$ .

To analyze the  $I$ -adic separation property of  $M$ , the following Artin-Rees Lemma is particularly useful.

**Theorem 1.7.4** (Artin-Rees Lemma). Let  $A$  be a noetherian ring,  $I$  an ideal of  $A$ ,  $M$  a finite  $A$ -module and  $N$  a submodule of  $M$ . Then there exists an integer  $r$  such that for all  $n \geq 0$ , we have

$$(I^{r+n}M) \cap N = I^n(I^rM \cap N).$$

*Proof.* Let

$$A' := A \oplus IX \oplus I^2X^2 \oplus \cdots \subset A[X]$$

be a graded  $A$ -algebra. Note that if  $I = (a_1, \dots, a_k)$ , then  $A' = A[a_1X, \dots, a_kX]$ . Hence  $A'$  is a noetherian ring. Let

$$M' := M \oplus IMX \oplus I^2MX^2 \oplus \cdots$$

be a graded  $A'$ -module. Then  $M'$  is a finite  $A'$ -module since it is generated by  $M$  and  $M$  is finite over  $A$ . Let

$$N' := N \oplus (IM \cap N)X \oplus (I^2M \cap N)X^2 \oplus \cdots$$

be a graded submodule of  $M'$ . Then  $N'$  is finite over  $A'$ . Suppose  $N' = \sum A'x_i$  with  $x_i \in I^{d_i}M \cap N$ . Choose  $r \geq d_i$  for all  $i$ . Then the degree  $n+r$  part of  $N'$  is equal to degree  $n$  part of  $A'$  timing the degree  $r$  part of  $N'$ . That is, for all  $n \geq 0$ ,  $I^{n+r}M \cap N = I^n(I^rM \cap N)$ .  $\square$

**Corollary 1.7.5.** Let  $A$  be a noetherian ring,  $I$  an ideal of  $A$ ,  $M$  a finite  $A$ -module and  $N$  a submodule of  $M$ . Then the subspace topology on  $N$  induced by  $N \subset M$  coincides with the  $I$ -adic topology on  $N$ .

*Proof.* This is a direct consequence of the Artin-Rees Lemma.  $\square$

**Corollary 1.7.6.** Let  $A$  be a noetherian ring,  $I$  an ideal of  $A$ , and  $M$  a finite  $A$ -module. Let  $N = \bigcap_{n \geq 0} I^n M$ . Then  $IN = N$ . In particular, if  $I \subset \text{rad}(A)$ , then  $M$  is  $I$ -adically separated.

*Proof.* We have that

$$N = I^{n+r}M \cap N = I^n(I^rM \cap N) = I^nN \subset IN \subset N.$$

The latter conclusion follows from the Nakayama's Lemma.  $\square$

**Definition 1.7.7.** Let  $A$  be a ring,  $I$  an ideal of  $A$  and  $M$  an  $A$ -module. We say that  $M$  is *complete* (with respect to  $I$ -adic topology) if  $M$  is  $I$ -adically separated and complete as a metric space with respect to the metric induced by the  $I$ -adic topology.

**Lemma 1.7.8.** Let  $A$  be a ring,  $I$  an ideal of  $A$  and  $M$  an  $A$ -module. Then the inverse limit

$$\hat{M} := \varprojlim (\cdots \rightarrow M/I^nM \rightarrow M/I^{n-1}M \rightarrow \cdots \rightarrow M/IM)$$

exists in the category of  $A$ -modules. Moreover,  $\hat{A}$  is an  $A$ -algebra and  $\hat{M}$  is an  $\hat{A}$ -module.

*Proof.* Let

$$\hat{M} := \left\{ (x_n) \in \prod_{n \geq 0} M/I^nM \mid x_{n+1} \mapsto x_n \right\}.$$

We claim that  $\hat{M}$  is that we desired. **Yang: To be completed.**  $\square$

**Definition 1.7.9** (Formal Completion). Let  $A$  be a ring,  $I$  an ideal of  $A$  and  $M$  an  $A$ -module. The *formal completion* of  $M$  with respect to  $I$ , denoted by  $\widehat{M}$ , is defined as

$$\widehat{M} := \varprojlim (\cdots \rightarrow M/I^n M \rightarrow M/I^{n-1} M \rightarrow \cdots \rightarrow M/IM),$$

where the maps are the natural projections  $M/I^n M \rightarrow M/I^{n-1} M$ .

**Example 1.7.10.** Let  $A = \mathbb{Z}$  be the ring of integers and  $I = p\mathbb{Z}$ . The formal completion of  $\mathbb{Z}$  with respect to  $p\mathbb{Z}$  is the ring of  $p$ -adic integers, denoted by  $\mathbb{Z}_p$ . The elements of  $\mathbb{Z}_p$  can be represented as infinite series of the form

$$a_0 + a_1 p + a_2 p^2 + \cdots,$$

where  $a_i \in \{0, 1, \dots, p-1\}$ .

**Example 1.7.11.** Let  $R$  be a ring,  $A = R[X_1, \dots, X_n]$  and  $I = (X_1, \dots, X_n)$ . The formal completion of  $A$  with respect to  $I$  is the ring of formal power series  $R[[X_1, \dots, X_n]]$ . The elements of  $R[[X_1, \dots, X_n]]$  can be represented as infinite series of the form

$$\sum_{i_1, \dots, i_n} a_{i_1, \dots, i_n} X_1^{i_1} \cdots X_n^{i_n},$$

where  $a_{i_1, \dots, i_n} \in R$  and the multi-index  $(i_1, \dots, i_n)$  runs over all non-negative integers.

**Proposition 1.7.12.** The formal completion  $\widehat{M}$  of a  $A$ -module  $M$  is complete, and image of  $M$  is dense in  $\widehat{M}$ . Moreover,  $\widehat{M}$  is uniquely characterized by above properties.

*Proof.* Yang: To be completed. □

By the universal property of the inverse limit, we get a covariant functor from the category of  $A$ -modules to the category of topological  $\widehat{A}$ -modules, which sends an  $A$ -module  $M$  to  $\widehat{M}$  and a morphism  $f : M \rightarrow N$  to the induced morphism  $\widehat{f} : \widehat{M} \rightarrow \widehat{N}$ .

**Lemma 1.7.13.** Let

$$0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$$

be an exact sequence of finite  $A$ -modules. Then the sequence of  $\widehat{A}$ -modules

$$0 \rightarrow \widehat{M}_1 \rightarrow \widehat{M}_2 \rightarrow \widehat{M}_3 \rightarrow 0$$

is still exact.

*Proof.* Yang: To be completed. □

**Proposition 1.7.14.** Let  $\widehat{A}$  be completion of a noetherian ring  $A$  with respect to an ideal  $I$  and  $M$  a finite  $A$ -module. Then the natural map  $M \otimes_A \widehat{A} \rightarrow \widehat{M}$  is an isomorphism.

*Proof.* Since  $A$  is noetherian and  $M$  is finite, we have an exact sequence

$$A^m \rightarrow A^n \rightarrow M \rightarrow 0.$$

By Lemma 1.7.13, we have an exact sequence

$$\widehat{A}^m \rightarrow \widehat{A}^n \rightarrow \widehat{M} \rightarrow 0.$$

On the other hand, we have

$$A^m \otimes_A \hat{A} \rightarrow A^n \otimes_A \hat{A} \rightarrow M \otimes_A \hat{A} \rightarrow 0$$

by right exactness of the tensor product. Since the inverse limit commutes with finite direct sums, we complete the proof by the Five Lemma.  $\square$

**Proposition 1.7.15.** Let  $A$  be a noetherian ring and  $I$  an ideal of  $A$ . Then the formal completion  $\hat{A}$  of  $A$  with respect to  $I$  is a flat  $A$ -module.

*Proof.* This is a direct consequence of Lemma 1.7.13 and Proposition 1.7.14.  $\square$

**Lemma 1.7.16.** Let  $\hat{A}$  be the formal completion of a noetherian ring  $A$  with respect to an ideal  $I$ . Suppose that  $I$  is generated by  $a_1, \dots, a_n$ . Then we have an isomorphism of topological rings

$$\hat{A} \cong A[[X_1, \dots, X_n]]/(X_1 - a_1, \dots, X_n - a_n).$$

*Proof.* Yang: To be completed.  $\square$

**Proposition 1.7.17.** Let  $A$  be a noetherian ring and  $I$  an ideal of  $A$ . Then the formal completion  $\hat{A}$  of  $A$  with respect to  $I$  is a noetherian ring.

*Proof.* Note that  $A[[X_1, \dots, X_n]]$  is noetherian by Hilbert's Basis Theorem. Then the conclusion follows from Lemma 1.7.16.  $\square$

**Proposition 1.7.18.** Let  $A$  be a noetherian ring and  $\mathfrak{m}$  a maximal ideal of  $A$ . Then the formal completion  $\hat{A}$  of  $A$  with respect to  $\mathfrak{m}$  is a local ring with maximal ideal  $\mathfrak{m}\hat{A}$ .

*Proof.* Yang: To be completed.  $\square$

## 1.7.2 Complete local rings

Let  $(A, \mathfrak{m}, \mathbf{k})$  be a noetherian complete local ring with respect to the  $\mathfrak{m}$ -adic topology. We say that  $A$  is of *equal characteristic* if  $\text{char } A = \text{char } \mathbf{k}$ , and of *mixed characteristic* if  $\text{char } A \neq \text{char } \mathbf{k}$ . In latter case,  $\text{char } \mathbf{k} = p$  and  $\text{char } A = 0$  or  $\text{char } A = p^k$ .

The goal of this subsection is the following structure theorem for noetherian complete local rings due to Cohen.

**Theorem 1.7.19** (Cohen Structure Theorem). Let  $(A, \mathfrak{m}, \mathbf{k})$  be a noetherian complete local ring of dimension  $d$ . Then

- (a)  $A$  is a quotient of a noetherian regular complete local ring;
- (b) if  $A$  is regular and of equal characteristic, then  $A \cong \mathbf{k}[[X_1, \dots, X_d]]$ ;
- (c) if  $A$  is regular, of mixed characteristic  $(0, p)$  and  $p \notin \mathfrak{m}^2$ , then  $A \cong D[[X_1, \dots, X_{d-1}]]$ , where  $(D, p, \mathbf{k})$  is a complete DVR;
- (d) if  $A$  is regular, of mixed characteristic  $(0, p)$  and  $p \in \mathfrak{m}^2$ , then  $A \cong D[[X_1, \dots, X_d]]/(f)$ , where  $(D, p, \mathbf{k})$  is a complete DVR and  $f$  a regular parameter.

To prove the Cohen Structure Theorem, we first list some preliminary results on complete local rings. They are independently important and can be used in other contexts.

**Theorem 1.7.20** (Hensel's Lemma). Let  $(A, \mathfrak{m}, \mathbf{k})$  be a complete local ring,  $f \in A[X]$  a monic polynomial and  $\bar{f} \in \mathbf{k}[X]$  its reduction modulo  $\mathfrak{m}$ . Suppose that  $\bar{f} = \bar{g} \cdot \bar{h}$  for some monic polynomials  $\bar{g}, \bar{h} \in \mathbf{k}[X]$  such that  $\gcd(\bar{g}, \bar{h}) = 1$ . Then the factorization lifts to a unique factorization  $f = g \cdot h$  in  $A[X]$  such that  $g$  and  $h$  are monic polynomials.

*Proof.* Lift  $\bar{g}$  and  $\bar{h}$  to monic polynomials  $g_1, h_1 \in A[X]$ . We inductively construct a sequence of monic polynomials  $g_n, h_n \in A[X]$  such that  $\Delta_n = f - g_n h_n \in \mathfrak{m}^n[X]$  and  $g_n - g_{n+1}, h_n - h_{n+1} \in \mathfrak{m}^n[X]$  for all  $n \geq 1$ . Suppose that  $g_n$  and  $h_n$  are constructed. Let  $g_{n+1} = g_n + \varepsilon_n$  and  $h_{n+1} = h_n + \eta_n$  for  $\varepsilon_n, \eta_n \in \mathfrak{m}^n[X]$ . Then we have

$$f - g_{n+1}h_{n+1} = \Delta_n - (\varepsilon_n h_n + \eta_n g_n) + \varepsilon_n \eta_n.$$

Hence we just need to choose  $\varepsilon_n$  and  $\eta_n$  such that

$$\varepsilon_n h_n + \eta_n g_n \equiv \Delta_n \pmod{\mathfrak{m}^{n+1}}, \quad \deg \varepsilon_n < \deg g_n, \quad \deg \eta_n < \deg h_n.$$

Since  $\gcd(\bar{g}, \bar{h}) = 1$ , there exist  $\bar{u}, \bar{v} \in \mathbf{k}[X]$  such that  $\bar{u}\bar{g} + \bar{v}\bar{h} = 1$  and  $\deg \bar{u} < \deg \bar{g}$ ,  $\deg \bar{v} < \deg \bar{h}$ . Lift  $\bar{u}$  and  $\bar{v}$  to  $u, v \in A[X]$  preserving the degrees. Then we have  $ug_n + vh_n \equiv 1 \pmod{\mathfrak{m}}$ . Let  $\varepsilon_n = u\Delta_n$  and  $\eta_n = v\Delta_n$ . Then we get the desired equation.  $\square$

**Proposition 1.7.21.** Let  $(A, \mathfrak{m}, \mathbf{k})$  be a noetherian complete local ring and  $M$  an  $A$ -module that is  $\mathfrak{m}$ -adically separated. Suppose  $\dim_{\mathbf{k}} M/\mathfrak{m}M < \infty$ . Then the basis of  $M \otimes_A \mathbf{k}$  as  $\mathbf{k}$ -vector space can be lifted to a generating set of  $M$  as an  $A$ -module.

*Proof.* Let  $t_1, \dots, t_n \in M$  such that their images in  $M/\mathfrak{m}M$  form a basis of  $M/\mathfrak{m}M$  as a  $\mathbf{k}$ -vector space. Then  $M = t_1A + \dots + t_nA + \mathfrak{m}M$ . For every  $x \in M$ , we can write

$$x = a_{0,1}t_1 + \dots + a_{0,n}t_n + m_1$$

for some  $a_{0,i} \in A$  and  $m_1 \in \mathfrak{m}M$ . Inductively, we have  $\mathfrak{m}^k M = t_1\mathfrak{m}^k + \dots + t_n\mathfrak{m}^k + \mathfrak{m}^{k+1}M$ . Suppose that we have constructed  $m_k \in \mathfrak{m}^k M$ . Then we can write

$$m_k = a_{k,1}t_1 + \dots + a_{k,n}t_n + m_{k+1}.$$

Note that  $\sum_{k \geq 0} a_{k,i}$  converges in  $A$ , denote its limit by  $a_i$ . Then we have

$$x - a_1t_1 - \dots - a_nt_n = \sum_{i=1}^n \sum_{r \geq k} a_{r,i}t_i + m_k \in \mathfrak{m}^k M$$

for all  $k$ . Since  $M$  is  $\mathfrak{m}$ -adically separated,  $x = a_1t_1 + \dots + a_nt_n$ . It follows that  $M = \sum A t_i$ .  $\square$

The key to prove the Cohen Structure Theorem is the existence of coefficient rings.

**Definition 1.7.22** (Coefficient rings). Let  $(A, \mathfrak{m}, \mathbf{k})$  be a noetherian complete local ring. When  $A$  is equal-characteristic, the coefficient ring (or coefficient field) is a homomorphism of rings

$\mathbf{k} \rightarrow A$  such that  $\mathbf{k} \rightarrow A \rightarrow A/\mathfrak{m}$  is an isomorphism.

When  $A$  is mixed-characteristic, the coefficient ring is a complete local ring  $(R, pR, \mathbf{k})$  with a local homomorphism of rings  $R \hookrightarrow A$  such that the induced homomorphism  $R/pR \rightarrow A/\mathfrak{m}$  is an isomorphism.

**Remark 1.7.23.** Recall that a homomorphism of local rings  $f : (A, \mathfrak{m}_A) \rightarrow (B, \mathfrak{m}_B)$  is said to be local if  $f^{-1}(\mathfrak{m}_B) = \mathfrak{m}_A$ .

**Theorem 1.7.24.** Every noetherian complete local ring  $(A, \mathfrak{m}, \mathbf{k})$  has a coefficient ring.

Assume the existence of coefficient rings, we can prove the Cohen Structure Theorem.

*Proof of Cohen Structure Theorem.* Let  $R$  be a coefficient ring of  $A$  and  $\mathfrak{m} = (f_1, \dots, f_d)$  a minimal generating set of  $\mathfrak{m}$ . Then we have a homomorphism of complete local rings

$$\Phi : R[[X_1, \dots, X_d]] \rightarrow A, \quad X_i \mapsto f_i.$$

Let  $\mathfrak{n}$  be the maximal ideal of  $R[[X_1, \dots, X_d]]$ . Then  $\mathfrak{n}A = \mathfrak{m}$ . By Proposition 1.7.21,  $A$  is generated by 1 as an  $R[[X_1, \dots, X_d]]$ -module. This implies that  $\Phi$  is surjective and (a) follows.

If  $A$  is regular of equal characteristic, then  $\mathfrak{m}$  is generated by a regular sequence. By consider the dimension of  $R[[X_1, \dots, X_d]]$  and  $A$ , we have that  $\Phi$  is an isomorphism. This proves (b).

Note that if  $A$  is regular of mixed characteristic  $(0, p)$  and  $p \notin \mathfrak{m}^2$ , then  $\mathfrak{m}$  is generated by  $p, f_1, \dots, f_{d-1}$ . Then consider the homomorphism of complete local rings

$$R[[X_1, \dots, X_{d-1}]] \rightarrow A, \quad X_i \mapsto f_i.$$

By the same argument as above, we have that it is an isomorphism. This proves (c).

For (d), we have that  $\ker \Phi$  is of height 1 by the dimension argument. Since regular local rings are UFDs, we can write  $\ker \Phi = (f)$  for some  $f \in R[[X_1, \dots, X_d]]$ . Then we finish.  $\square$

## Existence of coefficient rings

*Proof of Theorem 1.7.24 in characteristic 0.* Note that for any  $n \in \mathbb{Z}$ ,  $n \notin \mathfrak{m}$ . Hence  $\mathbb{Q} \subset A$ . Let  $\Sigma := \{\text{subfield in } A\}$  and  $K$  a maximal element in  $\Sigma$  with respect to the inclusion. The set  $\Sigma$  is non-empty since  $\mathbb{Q} \in \Sigma$ . By Zorn's Lemma,  $K$  exists. Then  $K$  is a subfield of  $\mathbf{k}$  by  $K \hookrightarrow A \twoheadrightarrow A/\mathfrak{m} \cong \mathbf{k}$ . We claim that  $K$  is a coefficient field of  $A$ .

Suppose there is  $\bar{t} \in \mathbf{k} \setminus K$ . If  $\bar{t}$  is transcendental over  $K$ , lift  $\bar{t}$  to an element  $t \in A$ . Then for any polynomial  $f \neq 0 \in K[T]$ , we have  $f(\bar{t}) \neq 0 \in \mathbf{k}$ . Hence  $f(t) \notin \mathfrak{m}$ . This implies that  $1/f(t) \in A$ , whence  $K(t) \subset A$ . This contradicts the maximality of  $K$ . If  $\bar{t}$  is algebraic over  $K$ , let  $f \in K[T]$  be the minimal polynomial of  $\bar{t}$ . Then  $f$  is irreducible in  $K[T]$  and  $f(\bar{t}) = 0$ . Regard  $f$  as a polynomial in  $A[T]$  by  $K \hookrightarrow A$ . Note that  $\text{char } A = 0$  implies that  $f$  is separable. By Hensel's Lemma (Theorem 1.7.20), we can lift the root  $\bar{t}$  to an element  $t \in A$  such that  $f(t) = 0$ . Then  $K(t)$  is a field extension of  $K$  and  $K(t) \subset A$ . This contradicts the maximality of  $K$  again.  $\square$

The same strategy does not work when  $\text{char } \mathbf{k} = p > 0$  since there might be inseparable extensions. To fix this, we need to introduce the notion of  $p$ -basis.



**Definition 1.7.25.** Let  $\mathbf{k}$  be a field of characteristic  $p$ . A finite set  $\{t_1, \dots, t_n\} \subset \mathbf{k} \setminus \mathbf{k}^p$  is called *p-independent* if  $[\mathbf{k}(t_1, \dots, t_n) : \mathbf{k}] = p^n$ . A set  $\Theta \subset \mathbf{k} \setminus \mathbf{k}^p$  is called a *p-independent* if its any finite subset is *p-independent*. A *p-basis* for  $\mathbf{k}$  is a maximal *p-independent* set  $\Theta \subset \mathbf{k} \setminus \mathbf{k}^p$ .

By definition, we have that  $\mathbf{k} = \mathbf{k}^p[\Theta]$  for any *p-basis*  $\Theta$  of  $\mathbf{k}$ . For any  $a \in \mathbf{k}$  and  $\theta \in \Theta$ , we can write  $a$  as a polynomial in  $\Theta$  with coefficients in  $\mathbf{k}^p$ . The degree of  $\theta$  in such polynomial representation is at most  $p - 1$ . Such polynomial representation is unique by definition of *p-independence*.

Applying the Frobenius map  $n$  times, we have that  $\mathbf{k}^{p^n} = \mathbf{k}^{p^{n+1}}[\Theta^{p^n}]$ . This follows that  $\mathbf{k} = \mathbf{k}^{p^n}[\Theta]$  for all  $n$ . Moreover, for any  $a \in \mathbf{k}$  and  $\theta \in \Theta$ , we can write  $a$  as a polynomial in  $\Theta$  with coefficients in  $\mathbf{k}^{p^n}$  and the degree of  $\theta$  is at most  $p^n - 1$ . Such polynomial representation is unique.

Let  $\mathbf{k}$  be a perfect field of characteristic  $p$ . If there is  $a \in \mathbf{k} \setminus \mathbf{k}^p$ , then  $\mathbf{k}(a^{1/p})/\mathbf{k}$  is an inseparable extension. This contradicts the perfectness of  $\mathbf{k}$ . Hence  $\mathbf{k} = \mathbf{k}^p$  and  $\mathbf{k}$  has no nonempty *p-basis*.

**Example 1.7.26.** Let  $\mathbf{k} = \mathbb{F}_p(t_1, \dots, t_n)$ . Then  $\mathbf{k}^p = \mathbb{F}_p(t_1^p, \dots, t_n^p)$ . The set  $\{t_1, \dots, t_n\}$  is a *p-basis* for  $\mathbf{k}$ .

*Proof of Theorem 1.7.24 in characteristic p.* Choose  $\Theta \subset A$  such that its image in  $A/\mathfrak{m}$  is a *p-basis* for  $\mathbf{k}$ . Let  $A_n := A^{p^n} = \{a^{p^n} : a \in A\}$  and  $K := \bigcap_{n \geq 0} (A_n[\Theta])$ . Then we claim that  $K$  is a coefficient field of  $A$ .

First we show that  $A_n[\Theta] \cap \mathfrak{m} \subset \mathfrak{m}^{p^n}$ . For every  $a \in A_n[\Theta]$ , if the degree of  $\theta$  in the polynomial representation of  $a$  is more than  $p^n - 1$ , we can write  $\theta^k = \theta^{ap^n} \cdot \theta^b$  for some  $b < p^n$ . Regard  $\theta^{ap^n} \in A^{p^n}$  as coefficients. Now assume that  $a \in A_n[\Theta] \cap \mathfrak{m}$ . Then consider the image of  $a$  in  $A/\mathfrak{m}$ . The image of  $a$  equals  $0$  implies every coefficient of  $a$  is in  $\mathfrak{m}$ . Such coefficients are of form  $b^{p^n}$  for some  $b \in A$ , whence  $b \in \mathfrak{m}$ . Hence  $a \in \mathfrak{m}^{p^n}$ . This implies that  $K \cap \mathfrak{m} = \bigcap_{n \geq 0} (A_n[\Theta] \cap \mathfrak{m}) \subset \bigcap_{n \geq 0} \mathfrak{m}^{p^n} = \{0\}$ . Then  $K$  is a field and hence a subfield of  $\mathbf{k}$ .

For any  $\bar{a} \in \mathbf{k}$ , note that  $\mathbf{k} = \mathbf{k}^p[\bar{\Theta}] = \mathbf{k}^{p^2}[\bar{\Theta}] = \dots = \mathbf{k}^{p^n}[\bar{\Theta}] = \dots$ . For every  $n$ , write

$$\bar{a} = \sum_{\mu_n} \bar{c}_{\mu_n}^{p^n} \mu_n =: P_{\bar{a},n}(\bar{c}_{\mu_n}),$$

where  $\mu_n$  runs over all monomials in  $\bar{\Theta}$  with degree at most  $p^n - 1$  and  $\bar{c}_{\mu_n} \in \mathbf{k}$ . We call this representation the *p<sup>n</sup>-development* of  $\bar{a}$  with respect to  $\bar{\Theta}$ . Plug the *p<sup>m</sup>-development* of  $c_{\mu_n}$  into  $P_{\bar{a},n}$ , we get the *p<sup>n+m</sup>-development* of  $\bar{a}$ . In formula, that is,

$$P_{\bar{a},n}(P_{\bar{a},m}(\bar{c}_{\mu_{n+m}})) = P_{\bar{a},n+m}(\bar{c}_{\mu_{n+m}}).$$

Lift  $\bar{c}_{\mu_n}$  to  $c_{\mu_n} \in A$  for all  $\mu_n$ . Let  $a_n := P_{\bar{a},n}(c_{\mu_n}) = \sum_{\mu_n} c_{\mu_n}^{p^n} \mu_n \in A_n[\Theta]$ . For  $m \geq n$ , we have  $a_n - a_m \in A_n[\Theta] \cap \mathfrak{m} \subset \mathfrak{m}^{p^n}$ . Hence  $a_n$  converges to an element  $a \in A$ . Now we show that  $a \in K$ . For every  $\mu_k$ , let  $b_{\mu_k,n} \in A$  be the element getting by plugging  $c_{\mu_{n+k}}$  into the  $P_{\bar{a},n}$ . Then  $b_{\mu_k,n}$  converges to an element  $b_{\mu_k} \in A$ . By construction, we have

$$a = \lim_{n \rightarrow \infty} P_{\bar{a},n+k}(c_{\mu_{n+k}}) = \lim_{n \rightarrow \infty} P_{\bar{a},k}(b_{\mu_k,n}) = P_{\bar{a}}(b_{\mu_k}) = \sum_{\mu_k} b_{\mu_k}^{p^k} \mu_k \in A_k[\Theta], \quad \forall k.$$

It follows that  $a \in K$ . □

**Lemma 1.7.27.** Let  $(A, \mathfrak{m}, \mathbf{k})$  be a noetherian complete local ring of mixed characteristic. Suppose that  $\mathfrak{m}^n = 0$  for some  $n \geq 1$ . Then there exists a complete local ring  $(R, pR, \mathbf{k})$  with  $R \subset A$ .

*Proof.* Fix a  $p$ -basis of  $\mathbf{k}$  and lift it to  $\Theta \subset R$ . Let  $q = p^{n-1}$  and

$$m := \{\theta_1^{k_1} \cdots \theta_d^{k_d} \mid \theta_i \in \Theta, k_i \leq q-1\}, \quad S := \left\{ \sum_{\mu \in m, \text{ finite}} a_\mu \mu \mid a_\mu \in R^q \right\}.$$

For any  $a, b \in A$ , we claim that  $a \equiv b \pmod{\mathfrak{m}}$  if and only if  $a^q \equiv b^q \pmod{\mathfrak{m}^n}$ . If  $a \equiv b \pmod{\mathfrak{m}}$ , write  $a = b + m$  for some  $m \in \mathfrak{m}$ . Then  $a^p = b^p + pb^{q-1}m + \cdots + m^q$ . Hence  $a^p \equiv b^p \pmod{\mathfrak{m}^2}$ . Inductively, we have  $a^q \equiv b^q \pmod{\mathfrak{m}^n}$ . Conversely, if  $a^q \equiv b^q \pmod{\mathfrak{m}^n}$ , then  $a^q - b^q \in \mathfrak{m}^n \subset \mathfrak{m}$ . Note that the Frobenius map  $x \mapsto x^q$  is injective on  $A/\mathfrak{m}$ . It follows that  $a \equiv b \pmod{\mathfrak{m}}$ . By the claim,  $S$  maps to  $\mathbf{k}^q[\Theta] = \mathbf{k}$  bijectively.

Let

$$R := S + pS + p^2S + \cdots + p^{n-1}S.$$

We claim that  $R$  is a subring of  $A$ . If so,  $R/pR \cong \mathbf{k}$  and we get a complete local ring  $(R, pR, \mathbf{k})$ .

Take  $a, b \in A$ . We have

$$a^q + b^q = (a + b)^q + pc \in A^q + pA.$$

Inductively, we have

$$a^q + b^q \in A^q + pA^q + \cdots + p^{n-1}A^q.$$

This implies that  $R$  is closed under addition. Note that  $\theta^a = \theta^{aq} \cdot \theta^b$  with  $b < q$ . Then for any  $\mu, \nu \in m$ , we have  $\mu\nu \in S$ . Hence  $R$  is closed under multiplication.  $\square$

**Lemma 1.7.28.** Let  $\mathbf{k}$  be a field of characteristic  $p$ . Then there exists a DVR  $(D, pD, \mathbf{k})$  of mixed characteristic  $(0, p)$ .

*Proof.* Fix a well order  $\leq$  on  $\mathbf{k}$  and for any  $a \in \mathbf{k}$ , set  $\mathbf{k}_a$  be the subfield of  $\mathbf{k}$  generated by all elements  $b \in \mathbf{k}$  such that  $b \leq a$ . Then  $\mathbf{k} = \bigcup_{a \in \mathbf{k}} \mathbf{k}_a$ . We construct DVRs  $D_a$  with residue field  $\mathbf{k}_a$  such that  $D_a \subset D_b$  for  $a \leq b$ . Begin from  $\mathbf{k}_0 = \mathbb{F}_p$  and let  $D_0 = \mathbb{Z}_{(p)}$ . Suppose that  $D_a$  is constructed for all  $a < b$ . If  $\mathbf{k}_b/\mathbf{k}_a$  is transcendental, then let  $D_b$  be the localization of  $D_a[b]$  at the prime ideal generated by  $p$ .

If  $\mathbf{k}_b/\mathbf{k}_a$  is algebraic, then let  $\bar{f} \in \mathbf{k}_a[T]$  be the monic minimal polynomial of  $b$ . Let  $\mathbf{K}_a = \text{Frac}(D_a)$  and  $K_b = \mathbf{K}_a[T]/(\bar{f})$ , where  $f$  is a monic lift of  $\bar{f}$  to  $D_a[T]$ . Note that  $f$  is irreducible since  $\bar{f}$  is irreducible. Let  $D_b$  be the integral closure of  $D_a$  in  $K_b$ . In general,  $D_b$  is a Dedekind domain. Consider the prime factorization  $pD_b = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_k^{e_k}$  in  $D_b$ . For every  $i$ ,  $D_b/\mathfrak{p}_i$  is a field extension of  $\mathbf{k}_a$  and  $\bar{f}$  has a root in  $D_b/\mathfrak{p}_i$ . Suppose  $\deg \bar{f} = \deg f = d$ . It follows that  $[(D_b/\mathfrak{p}_i) : \mathbf{k}_a] = d$ . Note that we have  $\sum_{i=1}^k e_i f_i = [K_b : \mathbf{K}_a] = d$ . Hence  $k = 1$  and  $e_1 = 1$ . It follows that  $pD_b$  is prime and  $D_b$  is a DVR with residue field  $\mathbf{k}_b$ .

Let  $D = \bigcup_{a \in \mathbf{k}} D_a$ . Then  $(D, pD, \mathbf{k})$  is the desired DVR.  $\square$

**Example 1.7.29.** Let  $\mathbf{k} = \mathbb{F}_p(t)$ . Then  $D = \mathbb{Z}[t]_{(p)}$  is a DVR satisfying the condition in Lemma 1.7.28.

Let  $\mathbf{k} = \overline{\mathbb{F}_p}$ . For any  $n \geq 1$ , let  $K_n = K_{n-1}(\zeta_{p^{n-1}})$  and  $K_0 = \mathbb{Q}$ . Let  $D_n := \mathcal{O}_{K_n, \mathfrak{p}_n}$  be the

localization of the ring of integers of  $K_n$  at the prime  $\mathfrak{p}_n$  lying above  $\mathfrak{p}_{n-1}$ . Then  $D := \bigcup_n D_n$  is a DVR with residue field  $\mathbf{k}$ .

**Lemma 1.7.30.** Given  $\mathbf{k}$  a field of characteristic  $p$ , there exists a unique complete local ring  $(R, pR, \mathbf{k})$  of mixed characteristic  $(p^n, p)$ .

*Proof.* The existence follows from Lemma 1.7.28. To show the uniqueness, suppose that  $(R', pR', \mathbf{k})$  is another complete local ring of mixed characteristic  $(p^n, p)$ . Fix a  $p$ -basis of  $\mathbf{k}$  and lift it to  $\Theta \subset R$  and  $\Theta' \subset R'$  relatively. Let  $q = p^{n-1}$  and

$$m := \{\theta_1^{k_1} \cdots \theta_d^{k_d} \mid \theta_i \in \Theta, k_i \leq q-1\}, \quad S := \left\{ \sum_{\mu \in m, \text{ finite}} a_\mu \mu \mid a_\mu \in R^q \right\}.$$

Define  $m', S'$  similarly with  $\Theta'$  and  $R'$ . Since  $S \rightarrow R \rightarrow \mathbf{k}$  and  $S' \rightarrow R' \rightarrow \mathbf{k}$  are bijections, we can define a bijective map  $\Phi : S \rightarrow S'$ .

Note that any element in  $S$  can be written as  $s + pr$  with  $s \in S$  and  $r \in R$  uniquely since  $S \rightarrow \mathbf{k}$  is bijective. Inductively, we can write any element in  $R$  as

$$r = s + ps_1 + p^2s_2 + \cdots + p^{n-1}s_{n-1},$$

where  $s_i \in S$ . The similarly for  $R'$ . Extend  $\Phi$  to  $R$  and we get a bijection between  $R$  and  $R'$ . Note that by construction,  $\Phi$  preserves addition and multiplication. Hence we get a ring isomorphism  $\Phi : R \rightarrow R'$ .  $\square$

*Proof of Theorem 1.7.24 in mixed characteristic.* Since  $A$  is complete, we have  $A = \varprojlim_n A/\mathfrak{m}^n$ . By Lemma 1.7.27, there is a complete local ring  $(R_n, pR_n, \mathbf{k})$  with  $R_n \subset A/\mathfrak{m}^n$ . By Lemma 1.7.30, such  $R_n$  is unique up to isomorphism. It follows that  $R_n \cong R_m/p^{k_n}$  for  $m \geq n$ . We get an inverse system

$$\cdots \rightarrow R_n \rightarrow R_{n-1} \rightarrow \cdots \rightarrow R_1 \cong \mathbf{k}.$$

Let  $R := \varprojlim_n R_n$ . Then  $(R, pR, \mathbf{k})$  is a complete local ring. The homomorphisms  $R_n \hookrightarrow A/\mathfrak{m}^n$  induce a homomorphism of complete local rings  $R \hookrightarrow A$ . This concludes the proof.  $\square$



# Chapter 2

## Homological Algebra

### 2.1 Complexes and Homology

**Definition 2.1.1.** Let  $A_\bullet$  and  $B_\bullet$  be two complexes in  $\mathcal{A}$  and  $\varphi_\bullet, \psi_\bullet : A_\bullet \rightarrow B_\bullet$  be two morphisms of complexes. A *homotopy* between  $\varphi_\bullet$  and  $\psi_\bullet$  is a collection of morphisms  $h_n : A_n \rightarrow B_{n-1}$  such that

$$\varphi_n - \psi_n = d_{B_{n+1}} \circ h_n + h_{n-1} \circ d_{A_n}.$$

In diagram, we have

$$\begin{array}{ccccccc} \cdots & \longrightarrow & A_{n+1} & \longrightarrow & A_n & \xrightarrow{d_{A_n}} & A_{n-1} \longrightarrow \cdots \\ & & \searrow h_n & & \downarrow \psi_n & & \swarrow \varphi_n \\ \cdots & \longrightarrow & B_{n+1} & \xrightarrow{d_{B_{n+1}}} & B_n & \longrightarrow & B_{n-1} \longrightarrow \cdots \\ & & & & \uparrow h_{n-1} & & \end{array}$$

### 2.2 Derived Functors

In this section, fix an abelian category  $\mathcal{A}$ .

#### 2.2.1 Resolution

**Definition 2.2.1** (Resolution). Let  $A \in \mathcal{A}$ . A *projective resolution* (resp. *flat resolution*, *free resolution*) of  $A$  is an exact sequence

$$\cdots \rightarrow P_n \rightarrow P_{n-1} \rightarrow \cdots \rightarrow P_1 \rightarrow P_0 \rightarrow A \rightarrow 0,$$

where each  $P_i$  is a projective (resp. flat, free) object in  $\mathcal{A}$ .

An *injective resolution* of  $A$  is an exact sequence

$$0 \rightarrow A \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \cdots \rightarrow I^n \rightarrow \cdots,$$

where each  $I^i$  is an injective object in  $\mathcal{A}$ .

**Proposition 2.2.2.** Let  $P_\bullet : \cdots \rightarrow P_1 \rightarrow P_0 \rightarrow A \rightarrow 0$  and  $Q_\bullet : \cdots \rightarrow Q_1 \rightarrow Q_0 \rightarrow B \rightarrow 0$  be complexes in  $\mathcal{A}$  such that  $P_i$  is projective and  $Q_\bullet$  is exact. Given a morphism  $f : A \rightarrow B$ , there exists a morphism of complexes  $f_\bullet : P_\bullet \rightarrow Q_\bullet$  such that  $f_0 = f$ . In particular, any two such morphism of complexes are homotopic.

Dually, let  $I^\bullet : 0 \rightarrow A \rightarrow I^0 \rightarrow I^1 \rightarrow \cdots$  and  $J^\bullet : 0 \rightarrow B \rightarrow J^0 \rightarrow J^1 \rightarrow \cdots$  be complexes in  $\mathcal{A}$  such that  $J^i$  is injective and  $I^\bullet$  is exact. Given a morphism  $f : A \rightarrow B$ , there exists a morphism of complexes  $f^\bullet : I^\bullet \rightarrow J^\bullet$  such that  $f^0 = f$ . In particular, any two such morphism of complexes are homotopic.

*Proof.* Yang: To be completed. □

**Definition 2.2.3.** For an object  $A \in \mathcal{A}$ , the *projective dimension* of  $A$ , denoted  $\text{proj. dim } A$ , is the smallest integer  $n$  such that there exists a projective resolution

$$0 \rightarrow P_n \rightarrow P_{n-1} \rightarrow \cdots \rightarrow P_1 \rightarrow P_0 \rightarrow A \rightarrow 0$$

of  $A$  of length  $n$ . If no such  $n$  exists, we set  $\text{proj. dim } A = \infty$ .

Dually, the *injective dimension* of  $A$ , denoted  $\text{inj. dim } A$ , is the smallest integer  $n$  such that there exists an injective resolution

$$0 \rightarrow A \rightarrow I^0 \rightarrow I^1 \rightarrow \cdots \rightarrow I^{n-1} \rightarrow I^n \rightarrow 0$$

of  $A$  of length  $n$ . If no such  $n$  exists, we set  $\text{inj. dim } A = \infty$ .

## 2.3 Applications to Commutative Algebra

### 2.3.1 Homological dimension

**Lemma 2.3.1.** Let  $A$  be a ring and  $M$  an  $A$ -module. Then

$$\sup_M \text{proj. dim } M = \sup_N \text{inj. dim } N.$$

*Proof.* Note that

$$\text{proj. dim } M \leq n$$

if and only if

$$\text{Ext}_{n+1}^A(M, N) = 0, \quad \forall N.$$

And this is equivalent to

$$\text{inj. dim } N \leq n.$$

□

**Remark 2.3.2.** In fact, for fix  $N$ , we have

$$\text{inj. dim } N \leq n$$

if and only if

$$\text{Ext}_{n+1}^A(A/I, N) = 0, \quad \forall I$$

By Lemma Yang: ?. Hence we have

$$\sup_{M \text{ finite}} \text{proj. dim } M = \sup_M \text{proj. dim } M = \sup_N \text{inj. dim } N.$$

**Definition 2.3.3.** Let  $A$  be a ring. The *homological dimension* of  $A$ , denoted  $\text{hl. dim } A$ , is defined as

$$\text{hl. dim } A := \sup_M \text{proj. dim } M = \sup_M \text{inj. dim } M.$$

**Lemma 2.3.4.** Let  $A$  be a noetherian ring,  $B$  a flat  $A$ -algebra and  $M$  a finite  $A$ -module. Then we have

$$\text{Ext}_A^i(M, N) \otimes B \cong \text{Ext}_B^i(M \otimes A, N \otimes A), \quad \forall N.$$

*Proof.* Yang: To be completed. □

**Proposition 2.3.5.** Let  $A$  be a noetherian ring. Then

$$\text{hl. dim } A = \sup_{\mathfrak{p} \in \text{Spec } A} \text{hl. dim } A_{\mathfrak{p}}.$$

*Proof.* Compute homological dimension of  $A$  using  $\text{Ext}_A^i(M, N)$  for finite  $M$ . The conclusion follows from Propostion 2.3.4. □

**Definition 2.3.6.** Let  $(A, \mathfrak{m}, \mathbf{k})$  be a noetherian local ring. We say that a homomorphism of  $A$ -modules  $f : M \rightarrow N$  is *minimal* if the induced map  $M \otimes \mathbf{k} \rightarrow N \otimes \mathbf{k}$  is an isomorphism. Equivalently,  $f$  is minimal if and only if  $f$  is surjective and  $\text{Ker } f \subset \mathfrak{m}M$ .

**Definition 2.3.7.** Let  $A$  be a noetherian local ring and  $M$  a finite  $A$ -module. A *minimal projective resolution* of  $M$  is a projective resolution

$$\cdots \rightarrow P_n \xrightarrow{d_n} P_{n-1} \xrightarrow{d_{n-1}} \cdots \rightarrow P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} M \rightarrow 0$$

such that each homomorphism  $P_i \rightarrow \text{Ker } d_{i-1}$  is minimal.

**Proposition 2.3.8.** Let  $(A, \mathfrak{m}, \mathbf{k})$  be a noetherian local ring and  $M$  a finite  $A$ -module. Then  $M$  has a minimal projective resolution. Moreover, any two minimal projective resolutions of  $M$  are isomorphic.

*Proof.* Suppose  $M \otimes_A \mathbf{k} = \bigoplus \mathbf{k} \cdot \bar{x}_i$ . Lift  $x_i$  to elements of  $M$ . Then we have a minimal homomorphism  $d_0 : \bigoplus A \cdot x_i \rightarrow M$ . Similarly choose minimal homomorphisms  $d_k : A^{n_i} \rightarrow \text{Ker } d_{i-1}$  for  $i = 1, 2, \dots$ . This gives a minimal projective resolution.

Suppose we have two minimal homomorphism  $f, g : A^n \rightarrow M$ . After tensoring with  $\mathbf{k}$ , we have isomorphisms between  $f \otimes \mathbf{k}$  and  $g \otimes \mathbf{k}$ . Lifting to  $A$ , we get an homomorphism  $\varphi : f \rightarrow g$ . Here homomorphism between  $f, g$  means a homomorphism  $A^n \rightarrow A^n$  such that  $f = g \circ \varphi$ . The homomorphism  $\varphi$  is represented by a matrix  $T$ . We have  $\det T \notin \mathfrak{m}$ , whence  $\varphi$  is an isomorphism. □

**Proposition 2.3.9.** Let  $L_\bullet \rightarrow M$  be a minimal projective resolution and  $P_\bullet$  be an arbitrary projective resolution of  $M$ . Then we have  $P_\bullet \cong L_\bullet \oplus P'_\bullet$  for some exact complexes  $P'_\bullet$ .

*Proof.* By Proposition 2.2.2, we have homomorphism

$$L_\bullet \xrightarrow{\varphi_\bullet} P_\bullet \xrightarrow{\psi_\bullet} L_\bullet.$$

between complexes. By Proposition 2.2.2 again,  $T_\bullet := \psi_\bullet \circ \varphi_\bullet$  is homotopic to the identity by  $h_\bullet$ . Suppose  $T_\bullet$  is represented by a matrix. Since  $L_\bullet$  is minimal, we have

$$(T - \text{id})(L_n) = (d_{n+1} \circ h_n + h_{n-1} \circ d_n)(L_n) \subset \mathfrak{m}L_n.$$

Then  $\det T \notin \mathfrak{m}$  and hence  $T_\bullet$  is an isomorphism. It follows that  $\psi_\bullet$  is surjective, whence it splits  $P_\bullet$  into a direct sum  $L \oplus P'_\bullet$  since  $L_\bullet$  is projective. By the Five Lemma, we see that  $P'_\bullet$  is exact.  $\square$

**Lemma 2.3.10.** Let  $(A, \mathfrak{m}, \mathbf{k})$  be a noetherian local ring and  $M$  a finite  $A$ -module. Then  $\text{proj. dim } M \leq n$  if and only if  $\text{Tor}_{n+1}^A(M, \mathbf{k}) = 0$ .

*Proof.* The necessity is clear. For the sufficiency, we have a minimal projective resolution

$$\cdots \rightarrow P_{n+1} \xrightarrow{d_{n+1}} P_n \xrightarrow{d_n} P_{n-1} \xrightarrow{d_{n-1}} \cdots \rightarrow P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} M \rightarrow 0.$$

Let  $C := \mathfrak{Z}d_n$ . Then we have

$$0 \rightarrow P_{n+1} \xrightarrow{d_{n+1}} P_n \xrightarrow{d_n} C \rightarrow 0.$$

Hence  $\text{Tor}_1^A(C, \mathbf{k}) \cong \text{Tor}_{n+1}^A(M, \mathbf{k}) = 0$ . Let  $K = \text{Ker } d_n$ . Then we have the short exact sequence

$$0 \rightarrow K \rightarrow P_n \rightarrow C \rightarrow 0.$$

Since  $\text{Tor}_1^A(C, \mathbf{k}) = 0$ , there is an exact sequence

$$0 \rightarrow K \otimes_A \mathbf{k} \rightarrow P_n \otimes_A \mathbf{k} \rightarrow C \otimes_A \mathbf{k} \rightarrow 0.$$

Since  $P_n \rightarrow C$  is minimal, we have  $K \otimes_A \mathbf{k} = 0$ . By the Nakayama's lemma,  $K = 0$ . This implies that  $\text{proj. dim } C \leq 0$  and hence  $\text{proj. dim } M \leq n$ .  $\square$

**Proposition 2.3.11.** Let  $(A, \mathfrak{m}, \mathbf{k})$  be a noetherian local ring. Then  $\text{hl. dim } A = \text{proj. dim } \mathbf{k}$  (finite or infinite).

*Proof.* The inequality  $\text{hl. dim } A \geq \text{proj. dim } \mathbf{k}$  is by definition. Conversely, we can compute  $\text{Tor}_{n+1}^A(M, \mathbf{k})$  using minimal projective resolution of  $\mathbf{k}$  for any finite  $A$ -module  $M$ . By Lemma 2.3.10, we have  $\text{proj. dim } M \leq n$  if and only if  $\text{Tor}_{n+1}^A(M, \mathbf{k}) = 0$ . This implies that  $\text{proj. dim } M \leq n$  for all finite  $A$ -modules  $M$  if  $\text{proj. dim } \mathbf{k} = n$ . By Remark 2.3.2, we have  $\text{hl. dim } A \leq n$ .  $\square$

**Proposition 2.3.12.** Let  $(A, \mathfrak{m})$  be a noetherian local ring and  $M$  a finite  $A$ -module. Let  $a \in \mathfrak{m}$  be an  $M$ -regular element. Then  $\text{proj. dim } M/aM = \text{proj. dim } M + 1$ . Here we set  $\infty + 1 = \infty$ .

*Proof.* We have an exact sequence

$$0 \rightarrow M \xrightarrow{*a} M \rightarrow M/aM \rightarrow 0.$$



Take the long exact sequence with respect to  $\text{Tor}(-, \mathbf{k})$ , we get

$$\cdots \rightarrow \text{Tor}_{i+1}^A(M, \mathbf{k}) \rightarrow \text{Tor}_{i+1}^A(M/aM, \mathbf{k}) \rightarrow \text{Tor}_i^A(M, \mathbf{k}) \xrightarrow{*a} \text{Tor}_i^A(M, \mathbf{k}) \rightarrow \cdots$$

Since the derived homomorphism of  $*a$  is zero, we have  $\text{Tor}_{i+1}^A(M/aM, \mathbf{k}) = 0$  if and only if  $\text{Tor}_i^A(M, \mathbf{k}) = 0$ . By Lemma 2.3.10, we have  $\text{proj. dim } M/aM = \text{proj. dim } M + 1$ .  $\square$

### 2.3.2 Depth and regularity by homological algebra

**Proposition 2.3.13.** Let  $(A, \mathfrak{m}, \mathbf{k})$  be a noetherian local ring and  $M$  a finite  $A$ -module. Then

$$\text{depth } M := \inf\{i : \text{Ext}_A^i(\mathbf{k}, M) \neq 0\}.$$

*Proof.* Let  $a \in \mathfrak{m}$  be  $M$ -regular and  $N = M/aM$ . Then we claim that

$$\inf\{i : \text{Ext}_A^i(\mathbf{k}, N) \neq 0\} = \inf\{i : \text{Ext}_A^i(\mathbf{k}, M) \neq 0\} - 1.$$

Indeed, we have an exact sequence

$$0 \rightarrow M \xrightarrow{a} M \rightarrow N \rightarrow 0.$$

It induces a long exact sequence

$$\cdots \rightarrow \text{Ext}_A^{i-1}(\mathbf{k}, M) \rightarrow \text{Ext}_A^{i-1}(\mathbf{k}, N) \rightarrow \text{Ext}_A^i(\mathbf{k}, M) \xrightarrow{\text{Ext}_A^i(\mathbf{k}, \text{Mult}_a)} \text{Ext}_A^i(\mathbf{k}, M) \rightarrow \cdots.$$

Note that  $a \in \mathfrak{m}$ , then  $\text{Ext}_A^i(\mathbf{k}, \text{Mult}_a) = 0$ . It follows that when  $\text{Ext}_A^{i-1}(\mathbf{k}, M) = 0$ , we have  $\text{Ext}_A^i(\mathbf{k}, N) = 0$  iff  $\text{Ext}_A^i(\mathbf{k}, M) = 0$ , whence the claim.

Let  $n = \inf\{i : \text{Ext}_A^i(\mathbf{k}, M) \neq 0\}$ . Induct on  $n$ . Suppose first  $n = 0$ . Since  $\mathbf{k}$  is a simple  $A$ -module, there is an injective homomorphism  $\mathbf{k} \rightarrow M$ . Then  $\mathfrak{m} \in \text{Ass } M$  and hence  $\text{depth } M = 0$ .

Suppose  $n > 0$ , let  $a_1, \dots, a_m \in \mathfrak{m}$  be any  $M$ -regular sequence. Using the claim inductively on  $M/(a_1, \dots, a_m)M$ , we have  $n \geq \text{depth}$ . If  $M$  has no regular element, then  $\mathfrak{m} \subset \bigcup_{\mathfrak{p} \in \text{Ass } M} \mathfrak{p}$ . Then  $\mathfrak{m} = \mathfrak{p}$  for some  $\mathfrak{p} \in \text{Ass } M$ . This show that we can find  $x \neq 0 \in M$  such that  $\mathfrak{p} = \text{Ann } x$ . It gives a homomorphism  $\mathbf{k} = A/\mathfrak{m} \rightarrow M$ . That is a contradiction and hence  $M$  has a regular element. Let  $a$  be  $M$ -regular and  $N = M/aM$ . Then  $\text{depth } N = n - 1$  by the claim and induction hypothesis. Hence we have  $\text{depth } M \geq n$ .  $\square$

**Lemma 2.3.14.** Let  $(A, \mathfrak{m}, \mathbf{k})$  be a noetherian local ring. Suppose we have exact sequences

$$0 \rightarrow A^{n_r} \xrightarrow{d_r} A^{n_{r-1}} \xrightarrow{d_{r-1}} \cdots \rightarrow A^{n_1} \xrightarrow{d_1} A^{n_0},$$

such that  $A^{n_i} \rightarrow \text{Ker } d_{i-1}$  is minimal for all  $i$ . Then  $\text{depth } A \geq r$ .

*Proof.* Since  $d_r$  is injective and its image is contained in  $\mathfrak{m}A^{n_{r-1}}$ , we can choose  $t \in \mathfrak{m}$  that is not a zero divisor. Denote the sequence by  $C_\bullet$ . Then we have a short exact sequence of complexes

$$0 \rightarrow C_\bullet \xrightarrow{*t} C_\bullet \rightarrow C_\bullet/tC_\bullet \rightarrow 0.$$

Consider the long exact sequence in homology

$$\cdots \rightarrow H_i(C_\bullet) \xrightarrow{*t} H_i(C_\bullet) \rightarrow H_i(C_\bullet/tC_\bullet) \rightarrow H_{i-1}(C_\bullet) \xrightarrow{*t} H_{i-1}(C_\bullet) \rightarrow \cdots.$$

Since  $C_\bullet$  is exact, we have  $H_i(C_\bullet) = 0$  for all  $i$ . In particular,  $H_i(C_\bullet/tC_\bullet) = 0$  for all  $i \geq 2$ . Inductively, we can choose a regular sequence of length  $r$  in  $\mathfrak{m}$ .  $\square$

**Lemma 2.3.15.** Let  $(A, \mathfrak{m}, \mathbf{k})$  be a noetherian local ring and  $M$  a finite  $A$ -module. Suppose there is an injective homomorphism  $\mathbf{k} \rightarrow M$ . Then  $\text{proj. dim } M \geq \dim_{\mathbf{k}} T_{A, \mathfrak{m}}$ .

*Proof.* Let  $x_1, \dots, x_n \in \mathfrak{m} \setminus \mathfrak{m}^2$  such that their images in  $\mathfrak{m}/\mathfrak{m}^2$  form a basis. Then we have a complex

$$K_\bullet := 0 \rightarrow \wedge^n A^{\oplus n} \xrightarrow{d_n} \wedge^{n-1} A^{\oplus n} \xrightarrow{d_{n-1}} \cdots \rightarrow \wedge^1 A^{\oplus n} \xrightarrow{d_1} \wedge^0 A^{\oplus n} \xrightarrow{d_0} \mathbf{k} \rightarrow 0,$$

where

$$d_r : \wedge^r A^{\oplus n} \rightarrow \wedge^{r-1} A^{\oplus n}, \quad e_{i_1} \wedge \cdots \wedge e_{i_r} \mapsto \sum_{k=1}^r (-1)^k x_{i_k} e_{i_1} \wedge \cdots \wedge \widehat{e_{i_k}} \wedge \cdots \wedge e_{i_r}.$$

Here  $\widehat{e_{i_k}}$  means that we omit the  $k$ -th element. Let  $P_\bullet \rightarrow M$  be the minimal projective resolution of  $M$ . Then we have a homomorphism of complexes

$$\varphi_\bullet : K_\bullet \rightarrow P_\bullet$$

induced by the injective homomorphism  $\mathbf{k} \rightarrow M$ .

We claim that  $\varphi_i$  is injective and splits  $P_i$  into a direct sum  $K_i \oplus F_i$  with  $F_i$  free for all  $i \geq 0$ . Since  $K_i$  and  $P_i$  are free, we just need to show that  $\varphi_i \otimes_A \text{id}_{\mathbf{k}}$  is injective. Induct on  $i$ . For  $i = 0$ , note that  $\mathbf{k} \rightarrow M \otimes_A \mathbf{k}$  is injective, by the commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{\quad} & \mathbf{k} \\ \varphi_0 \otimes_A \text{id}_{\mathbf{k}} \downarrow & & \downarrow \\ P_0 \otimes_A \mathbf{k} & \xrightarrow{\cong} & M \otimes_A \mathbf{k} \end{array},$$

the image of  $\varphi_0 \otimes_A \text{id}_{\mathbf{k}}$  is not zero in  $P_0 \otimes_A \mathbf{k}$ .

For  $i > 0$ , since  $K_{i-1}$  and  $P_{i-1}$  are free, we have a natural isomorphism between

$$\mathfrak{m}K_{i-1} \otimes_A \mathbf{k} \rightarrow \mathfrak{m}P_{i-1} \otimes_A \mathbf{k}$$

and

$$K_{i-1} \otimes_A \mathfrak{m}/\mathfrak{m}^2 \rightarrow P_{i-1} \otimes_A \mathfrak{m}/\mathfrak{m}^2.$$

We have a commutative diagram

$$\begin{array}{ccc} K_i \otimes_A \mathbf{k} & \longrightarrow & \mathfrak{m}K_{i-1} \otimes_A \mathbf{k} \\ \downarrow & & \downarrow \\ P_i \otimes_A \mathbf{k} & \longrightarrow & \mathfrak{m}P_{i-1} \otimes_A \mathbf{k} \end{array} \quad (2.1)$$

Since  $P_{i-1}/K_{i-1} \cong F_{i-1}$  is free, the right vertical map in (2.1) is injective. By construction of  $K_\bullet$ ,  $K_i \otimes_A \mathbf{k} \rightarrow \mathfrak{m}K_{i-1} \otimes_A \mathbf{k}$  is injective. Hence the left vertical map in (2.1) is injective. This completes the proof of the claim.

By the claim,  $P_i \neq 0$  for all  $i \leq n$  and the conclusion follows.  $\square$

**Proposition 2.3.16** (Auslander-Buchsbaum formula). Let  $A$  be a noetherian local ring and  $M$  a finite  $A$ -module. Suppose  $\text{proj. dim } M < \infty$ . Then  $\text{proj. dim } M = \text{depth } A - \text{depth } M$ .

*Proof.* We have a minimal projective resolution

$$0 \rightarrow A^{n_r} \rightarrow A^{n_{r-1}} \rightarrow \cdots \rightarrow A^{n_1} \rightarrow A^{n_0} \rightarrow M \rightarrow 0.$$

By Lemma 2.3.14, we have  $\text{depth } A \geq \text{proj. dim } M$ .

Induct on  $\text{depth } M$ . Suppose  $\text{depth } M = 0$ . Then by Proposition 2.3.13, we have  $\text{Hom}_A(\mathbf{k}, M) \neq 0$ , whence there is an injective homomorphism  $\mathbf{k} \rightarrow M$ . By Lemma 2.3.15, we have

$$\text{depth } A \geq \text{proj. dim } M \geq \dim_{\mathbf{k}} T_{A, \mathfrak{m}} \geq \text{depth } A.$$

If  $\text{depth } M > 0$ , choose a regular element  $a \in \mathfrak{m}$  that is  $M$ -regular. Then by Proposition 2.3.12, we have

$$\text{depth } M + \text{proj. dim } M = \text{depth}(M/aM) - 1 + \text{proj. dim}(M/aM) + 1 = \text{depth } A.$$

□

**Theorem 2.3.17.** Let  $(A, \mathfrak{m})$  be a noetherian local ring. Then  $A$  is regular at  $\mathfrak{m}$  if and only if  $\text{hl. dim } A < \infty$ .

*Proof.* Suppose  $A$  is regular at  $\mathfrak{m}$ . Let  $x_1, \dots, x_n$  be a minimal generating set of  $\mathfrak{m}$ . Then  $x_1, \dots, x_n$  is an  $A$ -regular sequence since  $A$  is regular at  $\mathfrak{m}$ . By Proposition 2.3.12, we have  $\text{proj. dim } \mathbf{k} = \text{proj. dim } A/(x_1, \dots, x_n)A = n + \text{proj. dim } A = n$ .

Conversely, suppose  $\text{hl. dim } A < \infty$ . Then by Proposition 2.3.11, we have  $\text{proj. dim } \mathbf{k} < \infty$ . We have

$$\dim_{\mathbf{k}} T_{A, \mathfrak{m}} \leq \text{proj. dim } \mathbf{k} \leq \text{depth } A \leq \dim_{\mathbf{k}} T_{A, \mathfrak{m}}.$$

The first " $\leq$ " follows from Lemma 2.3.15. The second " $\leq$ " follows from Proposition 2.3.16. Hence we see that  $A$  is regular at  $\mathfrak{m}$ . □

**Corollary 2.3.18.** Let  $(A, \mathfrak{m})$  be a noetherian local ring. Then  $A$  is regular if and only if it is regular at  $\mathfrak{m}$ .

*Proof.* The sufficiency is trivial. For the necessity, note that if  $A$  is regular, then  $\text{hl. dim } A < \infty$  by Theorem 2.3.17. For any  $\mathfrak{p} \in \text{Spec } A$ , we have a finite projective resolution

$$0 \rightarrow P_n \rightarrow P_{n-1} \rightarrow \cdots \rightarrow P_1 \rightarrow P_0 \rightarrow A/\mathfrak{p} \rightarrow 0.$$

Tensoring with  $A_{\mathfrak{p}}$ , we have a finite projective resolution of  $\kappa(\mathfrak{p})$ . By Theorem 2.3.17 again, we see that  $A_{\mathfrak{p}}$  is regular at  $\mathfrak{p}$ . □

**Lemma 2.3.19.** Let  $A$  be a noetherian integral domain. Then  $A$  is a UFD if and only if every height 1 prime ideal of  $A$  is principal.

*Proof.* Yang: To be completed. □

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**Lemma 2.3.20.** Let  $A$  be a noetherian integral domain and  $(x) \subset A$  a non-zero prime ideal. Then  $A$  is a UFD if and only if  $A[1/x]$  is a UFD.

| *Proof.* Yang: To be completed. □

**Theorem 2.3.21.** Let  $A, \mathfrak{m}$  be a regular noetherian local ring. Then  $A$  is UFD.

| *Proof.* Yang: To be completed. □

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# Chapter 3

## Category theory

### 3.1

### 3.2

### 3.3 Group objects

Let  $\mathbf{C}$  be a category with terminal object  $*$  and finite products.

**Definition 3.3.1.** A *group object* in  $\mathbf{C}$  is an object  $G$  together with morphisms

- $m : G \times G \rightarrow G$  (multiplication),
- $e : * \rightarrow G$  (identity),
- $i : G \rightarrow G$  (inverse),

such that the following diagrams commute:

- Associativity:

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{m \times \text{id}_G} & G \times G \\ \text{id}_G \times m \downarrow & & \downarrow m \\ G \times G & \xrightarrow{m} & G \end{array}$$

- Identity:

$$\begin{array}{ccccc} * \times G & \xrightarrow{e \times \text{id}_G} & G \times G & \xleftarrow{\text{id}_G \times e} & G \times * \\ & \searrow \cong & \downarrow m & \swarrow \cong & \\ & & G & & \end{array}$$

- Inverse:

$$\begin{array}{ccccc} G & \xrightarrow{\Delta} & G \times G & \xrightarrow{\text{id}_G \times i} & G \times G \xrightarrow{m} G \\ & \nwarrow e & & & \\ & & * & & \end{array}$$

Yang: To be checked.

**Example 3.3.2.** In the category of sets, a group object is just a group in the usual sense. In the category of topological spaces, a group object is a topological group. In the category of smooth manifolds, a group object is a Lie group. **Yang:**

**Definition 3.3.3.** Let  $G$  be a group object in a category  $\mathbf{C}$  and  $X$  an object in  $\mathbf{C}$ . A *group action* of  $G$  on  $X$  is a morphism

$$\sigma : G \times X \rightarrow X$$

such that the following diagrams commute:

- Identity:

$$\begin{array}{ccc} * \times X & \xrightarrow{e \times \text{id}_X} & G \times X \\ & \searrow \cong & \downarrow \sigma \\ & & X \end{array}$$

- Compatibility:

$$\begin{array}{ccc} G \times G \times X & \xrightarrow{m \times \text{id}_X} & G \times X \\ \text{id}_G \times \sigma \downarrow & & \downarrow \sigma \\ G \times X & \xrightarrow{\sigma} & X \end{array}$$

**Definition 3.3.4.** Let  $G$  be a group object in a category  $\mathbf{C}$  acting on objects  $X, Y$  via actions  $\sigma_X, \sigma_Y$  respectively. A morphism  $f : X \rightarrow Y$  is said to be  *$G$ -invariant* if the following diagram commutes:

$$\begin{array}{ccc} G \times X & \xrightarrow{\sigma_X} & X \\ \text{id}_G \times f \downarrow & & \downarrow f \\ G \times Y & \xrightarrow{\sigma_Y} & Y \end{array}$$

**Yang:**

**Definition 3.3.5.** Let  $G$  be a group object in a category  $\mathbf{C}$  acting on an object  $X$  via action  $\sigma$ . A *categorical quotient* of  $X$  by  $G$  is an object  $Q$  with trivial  $G$ -action, together with a  $G$ -invariant morphism  $q : X \rightarrow Q$ , such that for any object  $Y$  with trivial  $G$ -action and any  $G$ -invariant morphism  $f : X \rightarrow Y$ , there exists a unique morphism  $\bar{f} : Q \rightarrow Y$  making the following diagram commute:

$$\begin{array}{ccc} X & \xrightarrow{q} & Q \\ & \searrow f & \downarrow \bar{f} \\ & & Y \end{array}$$

**Yang:** Everything above need to be checked.

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# Chapter 4

## Site, sheaves and stacks





# Chapter 5

## Derived category

### 5.1 Definition and basic properties

#### 5.1.1 Preliminaries

Let  $\mathbf{A}$  be an abelian category. The category  $\mathbf{C}(\mathbf{A})$  of complexes in  $\mathbf{A}$  is defined as follows: the objects are complexes  $X^\bullet$  in  $\mathbf{A}$ , and the morphisms are morphisms of complexes. For every  $X^\bullet \in \text{Obj}(\mathbf{C}(\mathbf{A}))$ , the object  $X^n$  is the  $n$ -th component of the complex, and the morphism  $d^n : X^n \rightarrow X^{n+1}$  is the differential.

We denote  $X[k]$  by the complex obtained by shifting  $X^\bullet$  by  $k$ , that is,

$$X[k]^n = X^{n+k}, \quad d_{X[k]}^n = (-1)^k d_X^{n+k}.$$

Given a morphism  $f : X^\bullet \rightarrow Y^\bullet$  in  $\mathbf{C}(\mathbf{A})$ , we define the map cone  $\text{Cone}(f)^\bullet \in \mathbf{C}(\mathbf{A})$  by

$$\text{Cone}(f)^n = X^{n+1} \oplus Y^n, \quad d_{\text{Cone}(f)}^n = \begin{bmatrix} d_X^{n+1} & \\ f^{n+1} & d_Y^n \end{bmatrix},$$

Using the notation of shifting, we can also write

$$\text{Cone}(f)^\bullet = \left( X[1]^\bullet \oplus Y^\bullet, \begin{bmatrix} d_{X[1]} & \\ f[1] & d_Y \end{bmatrix} \right).$$

**Yang:** Check that the cone is a complex.

The category  $\mathbf{K}(\mathbf{A})$  is defined by

$$\text{Obj}(\mathbf{K}(\mathbf{A})) = \text{Obj}(\mathbf{C}(\mathbf{A})), \quad \text{Hom}_{\mathbf{K}(\mathbf{A})}(X^\bullet, Y^\bullet) = \text{Hom}_{\mathbf{C}(\mathbf{A})}(X^\bullet, Y^\bullet) / \{\text{homotopy}\}.$$

A homomorphism  $f^\bullet : X^\bullet \rightarrow Y^\bullet$  is said to be a *quasi-isomorphism* if the induced map  $H^n(f^\bullet) : H^n(X^\bullet) \rightarrow H^n(Y^\bullet)$  is an isomorphism for all  $n$ .

**Example 5.1.1.** Let  $\mathbf{A}$  be an abelian category and  $A$  an object in  $\mathbf{A}$ . Let  $A \xrightarrow{i} I^\bullet$  be an injective resolution of  $A$ . Then the complex  $I^\bullet$  is a complex in  $\mathbf{C}(\mathbf{A})$ , and the morphism

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & 0 & \longrightarrow & A & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & & & \downarrow i & & & & & & \\ \cdots & \longrightarrow & 0 & \longrightarrow & I^0 & \xrightarrow{d^0} & I^1 & \xrightarrow{d^1} & I^2 & \longrightarrow & \cdots \end{array}$$

is a quasi-isomorphism in  $\mathbf{K}(\mathbf{A})$ .

**Definition 5.1.2.** A *triangle* in  $\mathbf{K}(\mathbf{A})$  (or  $\mathbf{C}(\mathbf{A})$ ) is a diagram of the form

$$X^\bullet \xrightarrow{f} Y^\bullet \xrightarrow{g} Z^\bullet \xrightarrow{h} X[1]^\bullet$$

such that  $f$ ,  $g$ , and  $h$  are morphisms of complexes.

For every  $f^\bullet : X^\bullet \rightarrow Y^\bullet$  in  $\mathbf{C}(\mathbf{A})$ , we can construct a triangle

$$X^\bullet \xrightarrow{f^\bullet} Y^\bullet \rightarrow \text{Cone}(f^\bullet)^\bullet \rightarrow X[1]^\bullet,$$

where the morphism  $Y^\bullet \rightarrow \text{Cone}(f^\bullet)^\bullet$  is the natural inclusion, and the morphism  $\text{Cone}(f^\bullet)^\bullet \rightarrow X[1]^\bullet$  is the natural projection. The triangle which is isomorphic to the above triangle in  $\mathbf{K}(\mathbf{A})$  is called *distinguished triangle*.

**Definition 5.1.3** (Truncation functor). The *truncated functor*  $\tau^{>0} : \mathbf{K}(\mathbf{A}) \rightarrow \mathbf{K}(\mathbf{A})$  is defined by

$$\tau^{>0}(X^\bullet)^n = (\dots \rightarrow 0 \rightarrow \text{coker } d^0 \rightarrow X^1 \rightarrow X^2 \rightarrow \dots).$$

Yang: On cohomological level, we have

$$H^n(\tau^{>0}(X^\bullet)) = \begin{cases} 0, & n \leq 0, \\ H^n(X^\bullet), & n > 0. \end{cases}$$

**Definition 5.1.4** (Derived category). Let  $\mathbf{A}$  be an abelian category. The *derived category*  $\mathbf{D}(\mathbf{A})$  is defined by the following universal property: for any

**Proposition 5.1.5.** Let  $\mathbf{A}$  be an abelian category with enough injectives. Then for every object  $X \in \mathbf{D}^+(\mathbf{A})$ , there exists an isomorphism  $X \rightarrow I$  in  $\mathbf{D}^+(\mathbf{A})$  such  $I^n$  is an injective object in  $\mathbf{A}$  for all  $n$ .

**Definition 5.1.6.** Such an isomorphism  $X \rightarrow I$  is called an *injective resolution* of  $X$ .

**Definition 5.1.7** (Right Derived functor). Let  $\mathbf{A}$  and  $\mathbf{B}$  be abelian categories and  $F : \mathbf{A} \rightarrow \mathbf{B}$  a left exact functor. The *right derived functor* of  $F$  is a datum  $(T, \alpha)$  fitting into the following diagram

$$\begin{array}{ccc} \mathbf{K}^+(\mathbf{A}) & \xrightarrow{K^+(F)} & \mathbf{K}^+(\mathbf{B}) \\ \downarrow & \nearrow \alpha & \downarrow \\ \mathbf{D}^+(\mathbf{A}) & \xrightarrow{T} & \mathbf{D}^+(\mathbf{B}) \end{array}$$

that satisfies for every additive functor  $G : \mathbf{D}^+(\mathbf{A}) \rightarrow \mathbf{D}^+(\mathbf{B})$  preserving distinguish triangles and the

shifting  $X \mapsto X[1]$ , the map

$$\begin{array}{ccc}
 D^+(\mathbf{A}) & \xrightarrow{T} & D^+(\mathbf{B}) \\
 \Downarrow \beta & & \\
 D^+(\mathbf{A}) & \xrightarrow{G} & D^+(\mathbf{B})
 \end{array}
 \mapsto
 \begin{array}{ccc}
 K^+(\mathbf{A}) & \xrightarrow{K^+(F)} & K^+(\mathbf{B}) \rightarrow D^+(\mathbf{B}) \\
 & \Downarrow \alpha & \\
 K^+(\mathbf{A}) & \rightarrow D^+(\mathbf{A}) & \xrightarrow{T} D^+(\mathbf{B}) \\
 & \Downarrow \beta \circ_h \text{id} & \\
 K^+(\mathbf{A}) & \rightarrow D^+(\mathbf{A}) & \xrightarrow{G} D^+(\mathbf{B})
 \end{array}$$

is bijective.

Such functor is unique up to isomorphism, and denoted by  $RF$ .

**Proposition 5.1.8.** Let  $\mathbf{A}$  be an abelian category with enough injectives, and  $F : \mathbf{A} \rightarrow \mathbf{B}$  a left exact functor. Then the right derived functor  $RF$  is given by

$$RF(X^\bullet) = F(I^\bullet),$$

where  $I^\bullet$  is an injective resolution of  $X^\bullet$ .

### 5.1.2 An example

Fix a base ring  $T = \mathbb{Z}_p[[u]]$  for some prime  $p > 0$  and let  $\mathfrak{x} = (p, T)$  be the maximal ideal of  $T$ . Let  $Z = \mathbb{P}_T^1$  be the projective line over  $T$ . Choose a covering of  $Z$  by two affine open subschemes  $U_0 = \text{Spec}(T[v])$  and  $U_1 = \text{Spec}(T[1/v])$ . Let  $I = (p, T, v) \subset T[v]$  be the ideal of the closed point  $z \in U_0 \subset Z$ .

Let  $\pi : X = \text{Bl}_p Z \rightarrow Z$  be the blow-up of  $Z$  at the point  $z$ . We try to describe it explicitly. Consider the blow-up  $\text{Proj } T[v][pW, uW, vW]$  of  $U_0$  at the point  $z$ , where  $W$  is a formal variable to denote grading. It is covered by

$$\begin{aligned}
 U_{01} &= \text{Spec} \left( T[v] \left[ \frac{uW}{pW}, \frac{vW}{pW} \right] \right) \cong, \\
 U_{02} &= \text{Spec} \left( T[v] \left[ \frac{pW}{uW}, \frac{vW}{uW} \right] \right) \cong, \\
 U_{03} &= \text{Spec} \left( T[v] \left[ \frac{pW}{vW}, \frac{uW}{vW} \right] \right) \cong.
 \end{aligned}$$

Reduce to the special fiber, they become

$$\begin{aligned}
 U_{01,x} &= \text{Spec} \left( \mathbb{F}_p \left[ \frac{uW}{pW}, \frac{vW}{pW} \right] \right), \\
 U_{02,x} &= \text{Spec} \left( \mathbb{F}_p \left[ \frac{pW}{uW}, \frac{vW}{uW} \right] \right), \\
 U_{03,x} &= \text{Spec} \left( \mathbb{F}_p[v] \left[ \frac{pW}{vW}, \frac{uW}{vW} \right] / (v \frac{pW}{vW}, v \frac{uW}{vW}) \right) \cong \text{Spec} (\mathbb{F}_p[v, \alpha, \beta] / (v\alpha, v\beta)).
 \end{aligned}$$

Glue these three affine schemes and  $U_{1,x}$  together, we obtain the special fiber  $X_x$ , which consists of two components  $\mathbb{P}_{\mathbb{F}_p}^1$  and  $\mathbb{P}_{\mathbb{F}_p}^2$  meeting at one point. It follows that the exceptional divisor  $E$  of the blow-up  $\pi : X \rightarrow Z$  is isomorphic to  $\mathbb{P}_{\mathbb{F}_p}^2$ .

Reduce to the fiber  $p = 0$ , we have

$$\begin{aligned} U_{01,p} &= \operatorname{Spec} \left( \mathbb{F}_p \left[ \frac{uW}{pW}, \frac{vW}{pW} \right] \right), \\ U_{02,p} &= \operatorname{Spec} \left( \mathbb{F}_p[[u]] \left[ \frac{pW}{uW}, \frac{vW}{uW} \right] / \left( u \frac{pW}{uW} \right) \right), \\ U_{03,p} &= \operatorname{Spec} \left( \mathbb{F}_p[[u]] \left[ v, \frac{pW}{vW}, \frac{uW}{vW} \right] / \left( v \frac{pW}{vW}, v \frac{uW}{vW} - u \right) \right). \end{aligned}$$

Let  $\mathcal{L} := \pi^* \mathcal{O}_Z(1)$  be the pullback of the line bundle  $\mathcal{O}_Z(1)$  on  $Z$ . **Yang: Then  $\mathcal{L}$  is nef and big.** We use this example to compute

$$R\Gamma_x(R\Gamma(X_{p=0}, \mathcal{L}))$$

**Definition 5.1.9.** Let  $X$  be a scheme and  $\mathcal{F}$  a coherent sheaf on  $X$ . For  $s \in \Gamma(X, \mathcal{F})$ , we define the *support* of  $s$  to be the closed subset  $\{x \in X \mid s_x \neq 0\}$ . Let  $Y \subset X$  be a closed subset. The section with support in  $Y$  is defined to be the set

$$\Gamma_Y(X, \mathcal{F}) = \{s \in \Gamma(X, \mathcal{F}) \mid \operatorname{Supp} s \subset Y\}.$$

Compute  $R\Gamma_x(R\Gamma(Z_{p=0}, \mathcal{O}_Z(1)))$